Marketing Analytics

Chapter 5 · Learning from Observational Data

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Part 1 · Observational data I

Before we start

Paper quiz on Chapter 4 *(10 min)*

- Individual, closed book, counts toward participation
- Afterwards: a few leftovers from Chapter 4 and a look at the quiz questions

Assignment 2 is running: deadline Group A Sun, Nov 8 · Group B Wed, Nov 11; debrief at the start of next session.

Devices only when we code

- **Input blocks:** laptops closed, phones away
- **Coding block** (40 min, at the end): laptops open, alone or with your neighbor; results via Menti, then two tables present

> Today: correlation and the regression line → reading a regression → adjustment and exogeneity → categorical predictors → prediction vs. explanation → **one coding block with two cases** at the end.

Today’s session

Observational data I *(today)*

5.1 Smart Mix without randomization
5.2 Correlation, the line, and least squares
5.3 Reading a regression
5.4 Adjusting for confounders — and exogeneity
5.5 Categorical predictors
5.6 Prediction vs. explanation

Observational data II *(next session)*

5.7 Functional form and elasticities
5.8 Logistic regression
5.9 Diagnostics that change decisions
5.10 Difference-in-differences

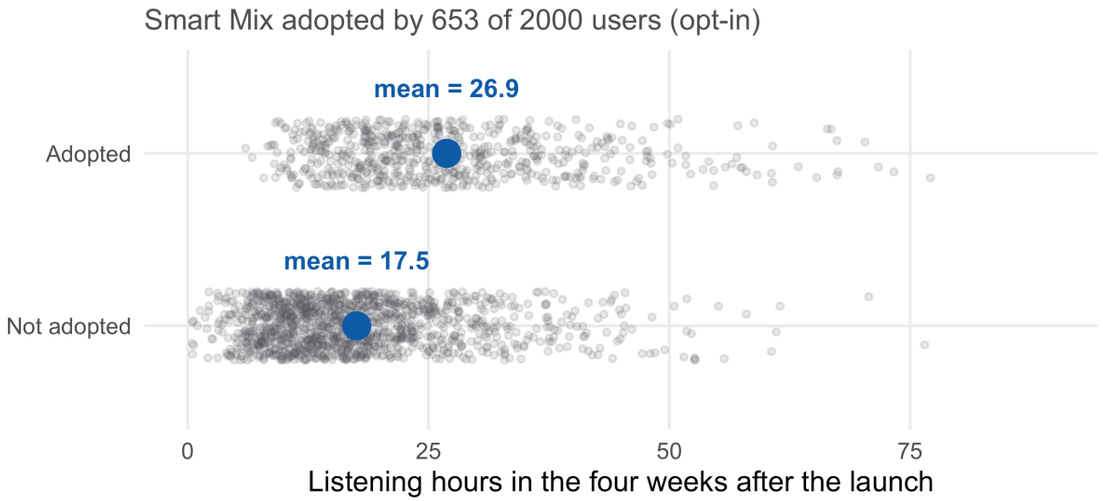
> **Driving question:** *What if we cannot randomize?*

5.1 Smart Mix without randomization

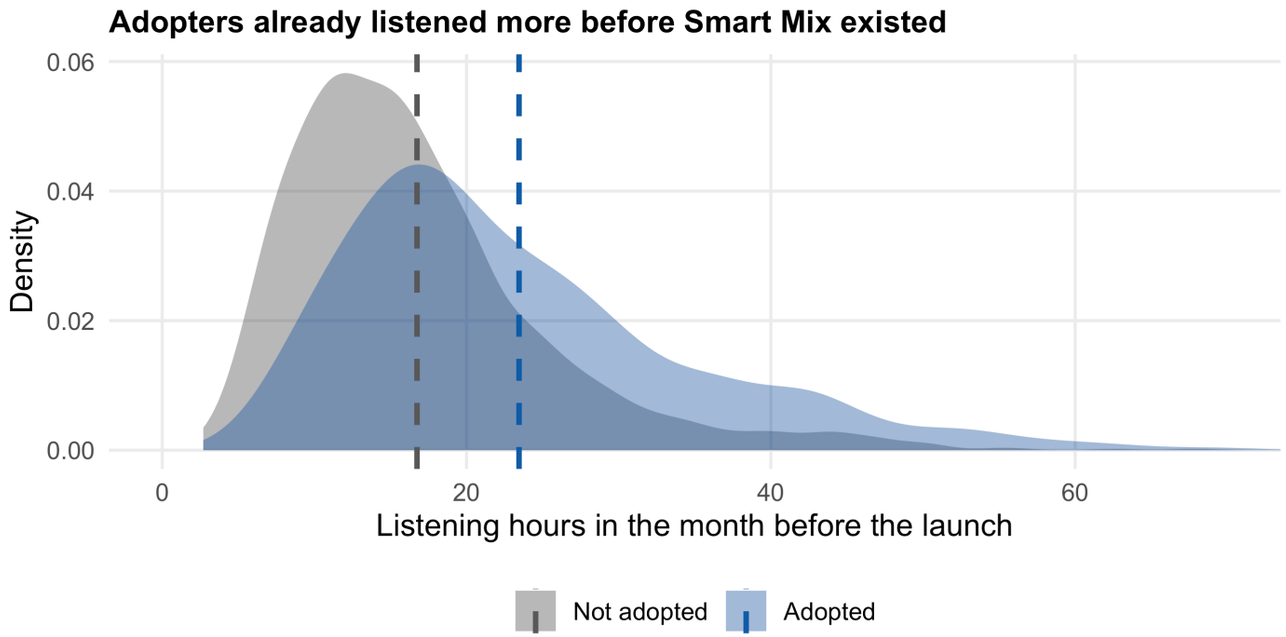
A different history

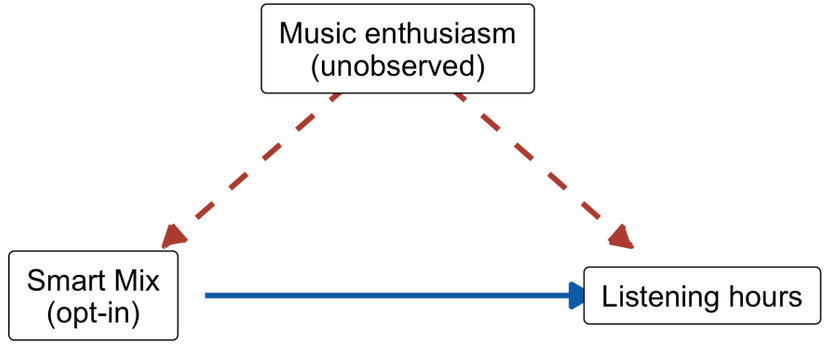
- Chapter 4: A/B test, 400 users → **+3.2 hours**
- Now: no experiment. Smart Mix launched as an **opt-in** feature
- Four weeks later: data of **2,000 users**, 653 adopters

The dashboard says: adopters listen **9.3 hours more** — three times the experimental effect. Would you report this number to management?



Adopters were different before Smart Mix existed

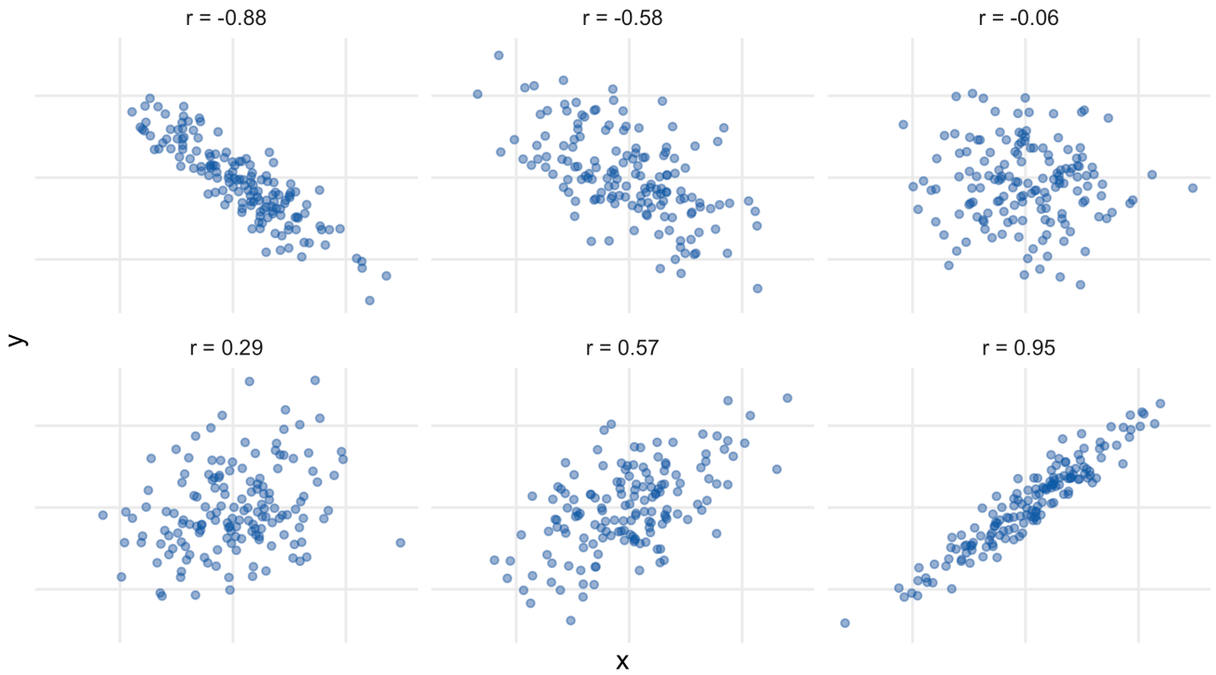




Same costume, new villain: **music enthusiasm** drives adoption *and* listening. Without a coin flip, we have one tool left: **adjustment**.

5.2 From correlation to a line

The correlation coefficient



$$r = \frac{\sum_i (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_i (x_i - \bar{x})^2}\sqrt{\sum_i (y_i - \bar{y})^2}}$$

- Standardized covariance → always between **−1 and +1**, no units
- **Sign** = direction · **size** = how tightly the cloud hugs a straight line
- Rules of thumb: 0.1 small · 0.3 medium · 0.5 large
- Smart Mix: `cor(prior_hours, hours)` = **0.64**

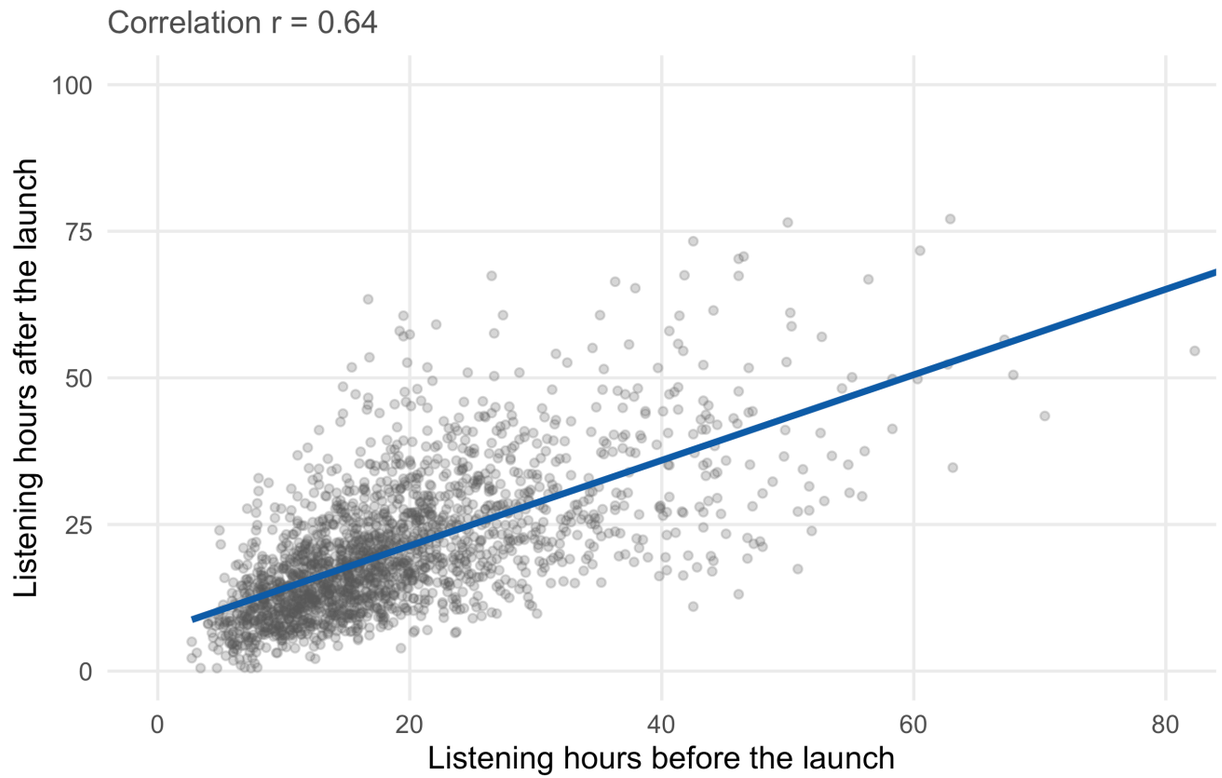
Two caveats: *r* measures **linear** association only (a U-shape can have $r \approx 0$) — and it is **symmetric**: nothing about which variable causes which.

Correlation vs. regression

	Correlation	Regression
Estimated coefficient	r, bounded between −1 and +1, no units	β_1, a priori not bounded, in units of y per unit of x
Interpretation	Linear association between two variables; **bidirectional** (r of x and y = r of y and x)	(Linear) relation of one or more predictors to an outcome; **directional** (x explains y)
Role of theory	Neither required nor testable	Required (which variable is the outcome, which predictors belong in the model) — and testable

> Correlation: *that* and *how tightly* two variables move together. Regression: *how much* y changes with x — which presupposes a direction, and that is where theory and this chapter's assumptions come in.

Two variables, one line



$$\text{hours}_i = \beta_0 + \beta_1 \,\text{prior hours}_i + \varepsilon_i$$

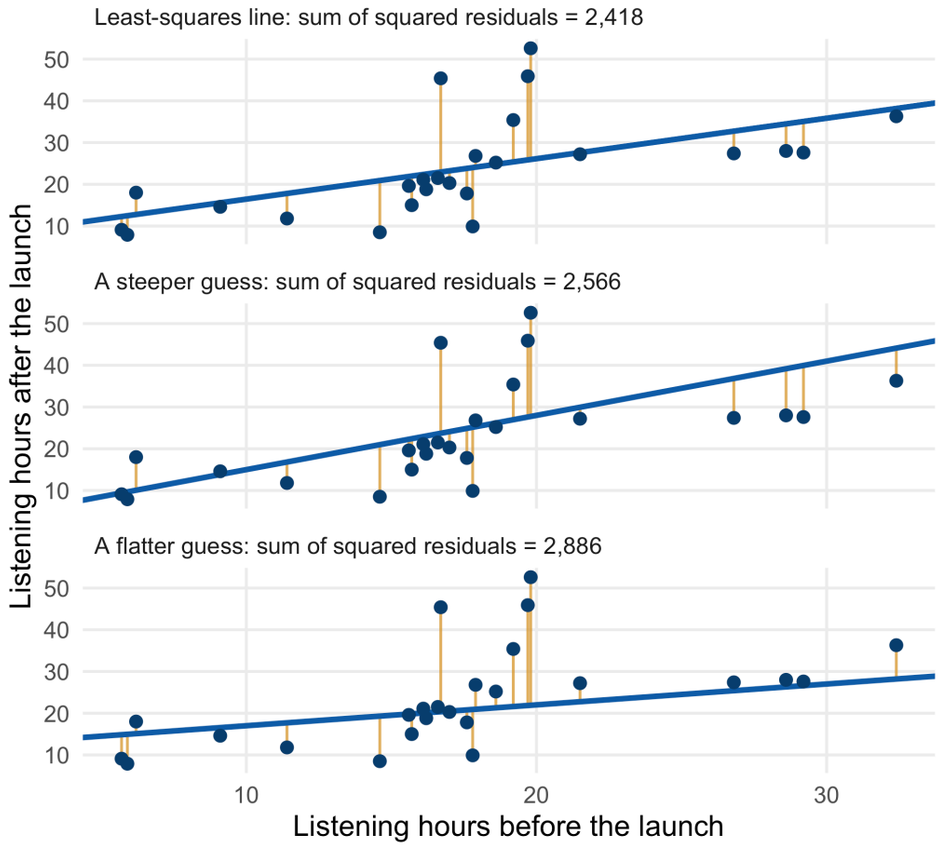
- i = one user · β_0 **intercept** · β_1 **slope** · ε_i **error term**
- Correlation: *that* they move together · regression: *how much*

```
coef(lm(hours ~ prior_hours, data = smartmix_obs))
```

```
(Intercept) prior_hours
     6.7849      0.7291
```

> Each extra hour before the launch **goes with** 0.73 hours after it — *goes with*, not *causes*.

Which line? Least squares



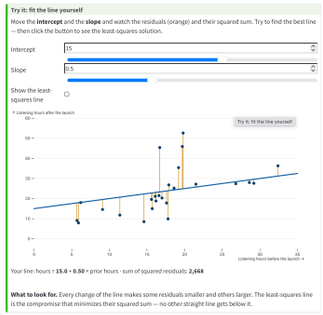
- Residual = observed − predicted (orange segments)
- **OLS** = the line that **minimizes the sum of squared residuals:**

$$\min_{\hat{\beta}_0, \hat{\beta}_1} \sum_{i=1}^{n} \left(y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i \right)^2$$

- Squaring: big misses count a lot; above and below alike · `lm()` does exactly this

Live demo: fit the line yourself — course book, §5.2

https://wu-rds.github.io/MA2026/05-observational-data.html#sec-obs-line



5.3 Reading a regression

The full output

```r
model_naive <- lm(hours ~ smart_mix, data = smartmix_obs)
summary(model_naive)
```

```
Call:
lm(formula = hours ~ smart_mix, data = smartmix_obs)

Residuals:
   Min     1Q Median     3Q    Max
-20.88  -7.64  -2.06   4.76 101.22

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)   17.541      0.300    58.5   <2e-16 ***
smart_mix      9.341      0.525    17.8   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 11 on 1998 degrees of freedom
Multiple R-squared:  0.137, Adjusted R-squared:  0.136
F-statistic:  316 on 1 and 1998 DF,  p-value: <2e-16
```

Reading the coefficients

`(Intercept)` **= 17.5**: the predicted hours when all predictors are 0 — here the mean of the **non-adopters**.

`smart_mix` **= 9.3**: the change in hours for a **one-unit increase** in the predictor. For a 0/1 variable that is the step from 0 to 1: **adopters − non-adopters**.

Continuous predictor (previous section): `prior_hours` = 0.73 → one more hour before the launch goes with **0.73 more hours** after it.

Several predictors (next section): every coefficient is read **"holding the other variables constant"**.

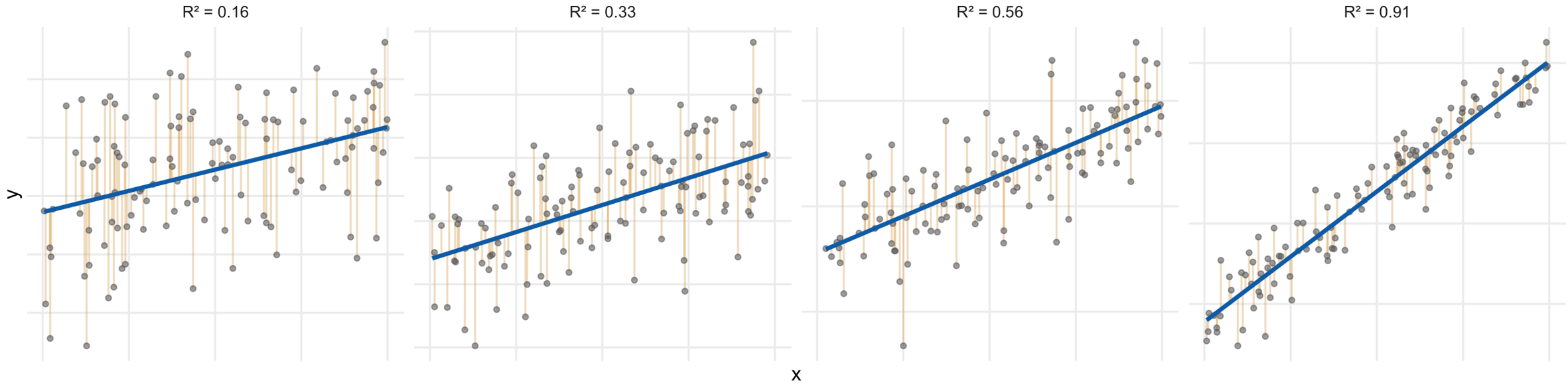
The general rule: a coefficient is the change in y when its predictor increases by **one unit** and nothing else changes. Whether that change is *causal* is the question of §5.4.

Line by line

Output	Meaning
`Estimate`	Intercept = mean of non-adopters (18.4); `smart_mix` = difference adopters − non-adopters (9.3)
`Std. Error`	Uncertainty of each coefficient (Chapter 3)
`t value`, `Pr(>|t|)`	*t*-test of each coefficient against zero (Chapter 4): signal / noise
`Residual standard error`	Typical distance of a user from the line, in hours
`Multiple R-squared`	Share of the variation in hours explained by the model (14%)
`F-statistic`	The whole model against "no predictors"

> With one 0/1 predictor, the regression **is** the comparison of two means — and its *t*-test is the two-sample *t*-test. `confint(model_naive)` gives the confidence intervals.

R^2: the share of explained variation



- Same idea as η^2 in the ANOVA: $R^2 = SS_M / SS_T$ — the share of the variation in y that the line explains
- Smart Mix dummy: 0.14 · prior listening: 0.41

> R^2 says how well the line **fits** — nothing about whether the slope is **causal**, and a small R^2 does not make an effect unimportant (14% explained, but 9 hours).

5.4 Adjusting for confounders

Holding prior listening constant

```r
model_prior <- lm(hours ~ smart_mix + prior_hours, data = smartmix_obs)
round(coef(summary(model_prior)), 3)
```

```
            Estimate Std. Error t value Pr(>|t|)
(Intercept)    6.451      0.412   15.65        0
smart_mix      4.901      0.443   11.07        0
prior_hours    0.662      0.020   33.17        0
```

> The Smart Mix coefficient drops from **9.3** to **4.9**: now the difference between adopters and non-adopters **with the same prior listening**. Part of the naive gap belonged to prior listening — and is now attributed to it.

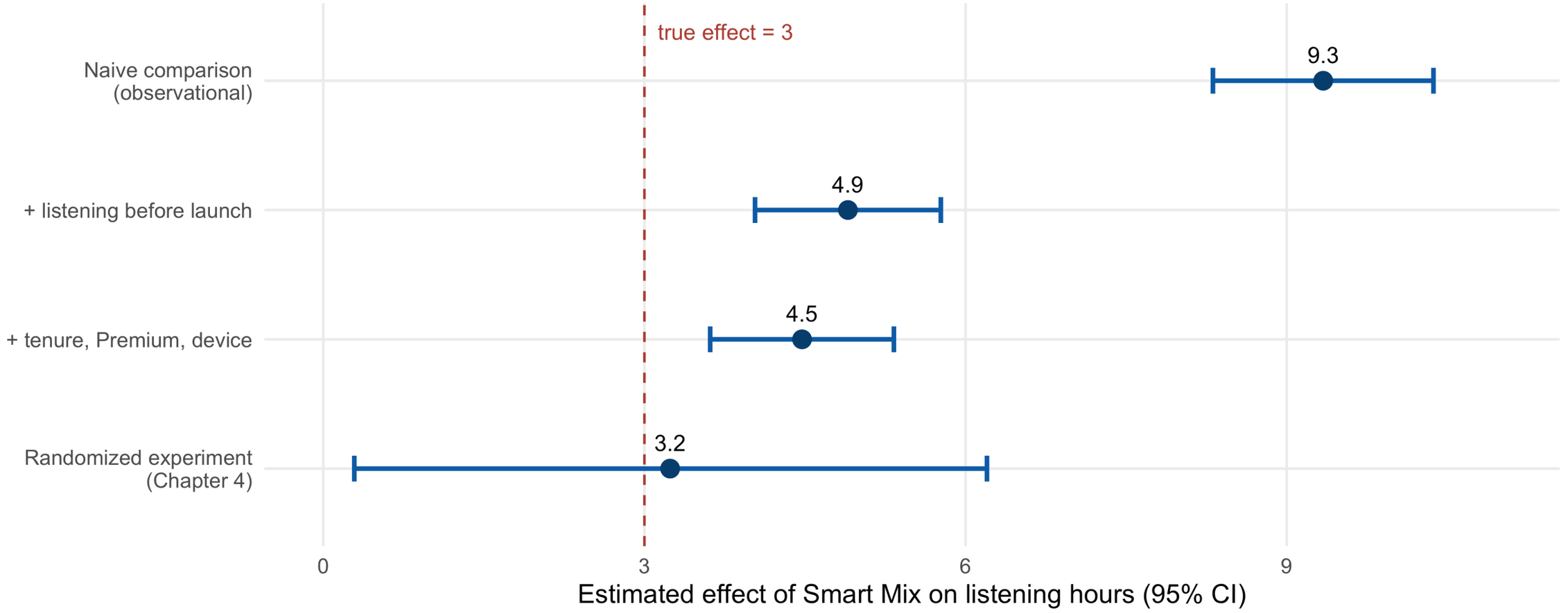
More variables, same logic

```
model_full <- lm(hours ~ smart_mix + prior_hours + tenure + premium + device, data = smartmix_obs)
round(coef(summary(model_full)), 3)
```

	Estimate	Std. Error	t value	Pr(>|t|)
(Intercept)	6.255	0.540	11.575	0.000
smart_mix	4.472	0.437	10.223	0.000
prior_hours	0.613	0.021	29.688	0.000
tenure	0.038	0.011	3.344	0.001
premium	2.924	0.426	6.869	0.000
deviceAndroid	−1.475	0.422	−3.493	0.000
deviceDesktop	−2.488	0.576	−4.320	0.000

Reading the table, one line each, "holding the other variables constant": `smart_mix` 4.5 = adopters listen 4.5 h more than non-adopters with the same prior listening, tenure, plan and device · `prior_hours` 0.61 = one more hour before the launch goes with 0.61 h more after it · `tenure` 0.04 = each month of tenure, +0.04 h · `premium` 2.9 = Premium users listen 2.9 h more than free users · `deviceAndroid` / `deviceDesktop` = difference to **iOS** (the reference). Tenure, Premium and device change little for Smart Mix: ≈ **4.5 hours**.

Naive, adjusted — and the truth

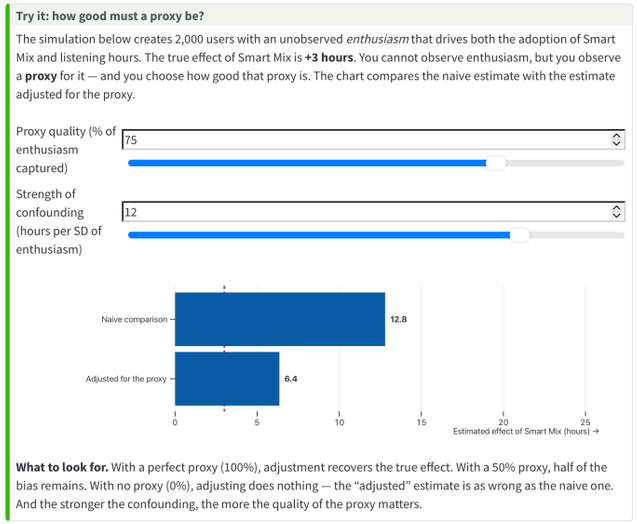


Adjustment removed ~75% of the bias — **not all of it**. We adjusted for a *proxy* of enthusiasm, not for enthusiasm. Regression removes exactly the confounding your variables capture.

Live demo: how good must a proxy be?

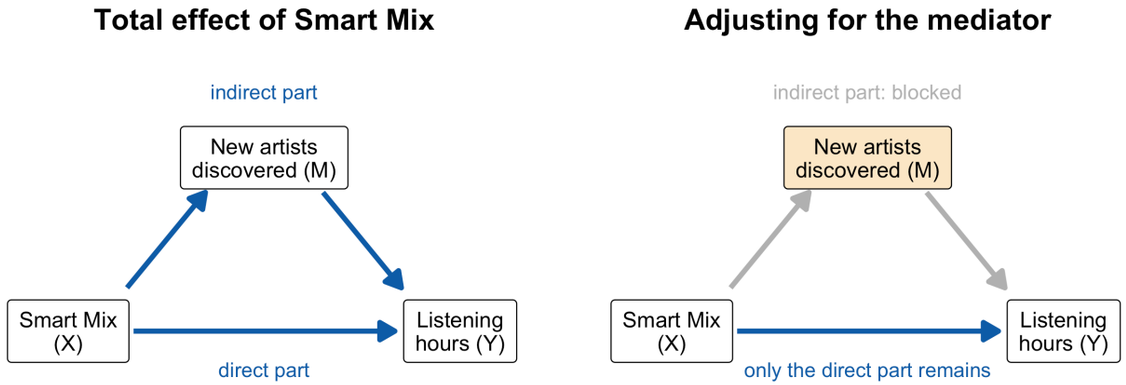
Course book, §5.4 — set the **quality of the proxy** and the **strength of confounding**.

https://wu-rds.github.io/MA2026/05-observational-data.html#sec-obs-adjust



> With a 50% proxy, how much of the bias is left? And what happens at 0%?

Two warnings



"Enough" is untestable. The assumption of no unobserved confounders cannot be checked in the data. Report *what* you adjusted for — and what you plausibly could not.

Adjusting for the wrong variable makes it worse. A **mediator** lies on the causal path: if Smart Mix works by making users discover new artists, adjusting for `new_artists` **blocks the indirect part** and the coefficient shows only the direct part. Only **confounders** — determined *before* the treatment, affecting treatment *and* outcome — belong in the model.

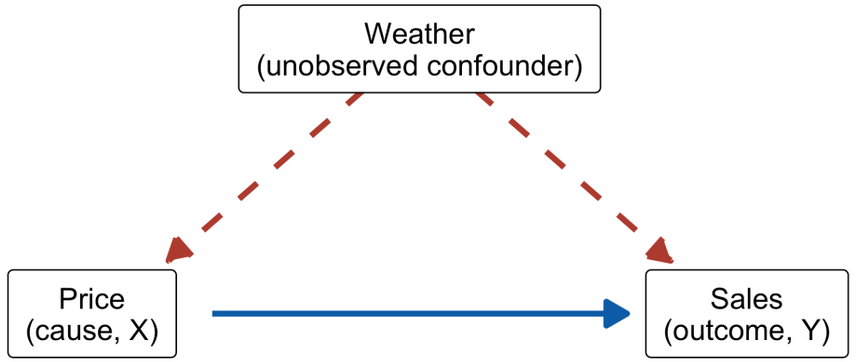
Where the unobserved variable goes

The true model (price and sales; weather raises both):

$$\text{Sales}_i = \beta_0 + \beta_1 \, \text{Price}_i + \beta_2 \, \text{Weather}_i + \phi_i$$

Weather is not in the data — so we estimate:

$$\text{Sales}_i = \beta_0 + \beta_1 \, \text{Price}_i + \underbrace{\varepsilon_i}_{= \beta_2 \text{Weather}_i + \phi_i}$$



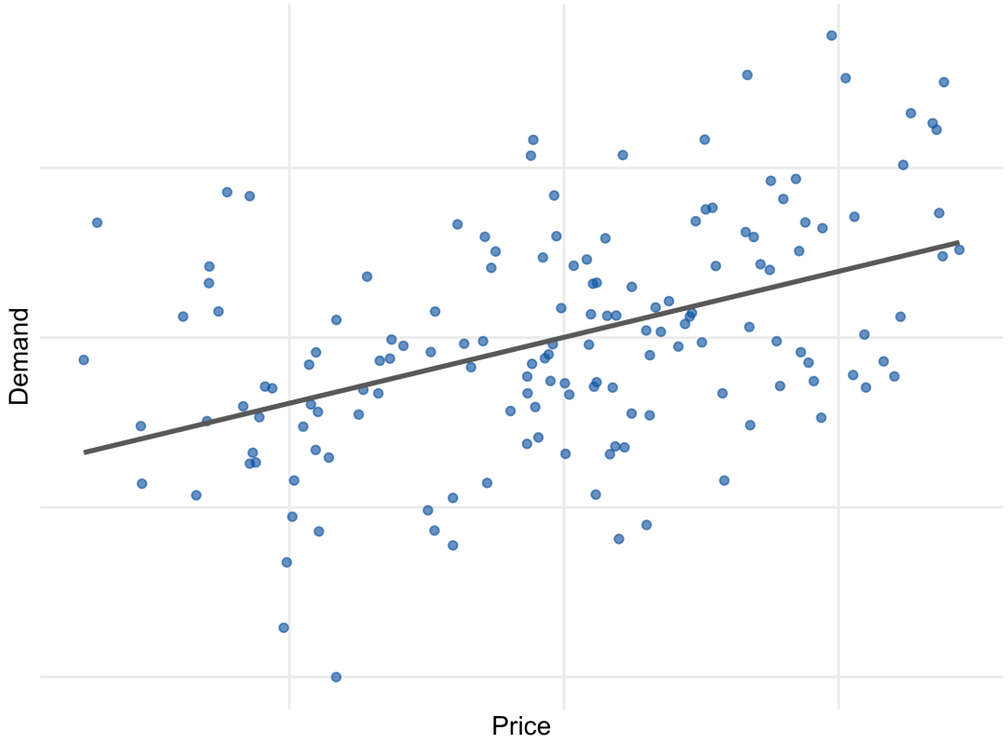
- The omitted variable does not disappear — it moves **into the error term**
- Weather also drives price → the error term is **correlated with the regressor**
- OLS attributes the weather effect to price: $\hat{\beta}_1$ is **biased**, and more data do not help

Smart Mix: replace price by **adoption**, weather by **enthusiasm**. Enthusiasm is in ε_i and drives adoption → 9.3 h instead of 3.

Variation has two components

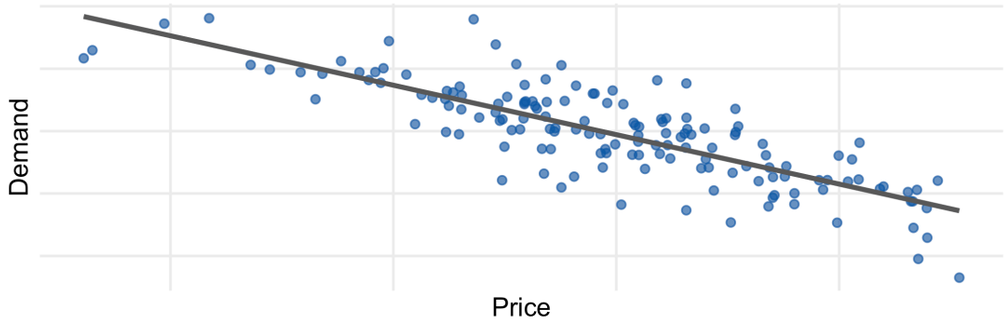
Observed variation

The data: higher prices, higher demand?



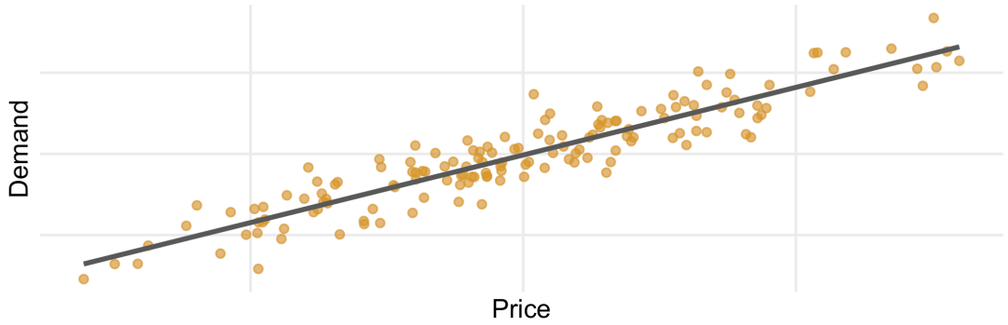
Exogenous variation

Price moves for its own reasons: the causal effect



Endogenous variation

Weather moves price AND demand: not the effect



The observed price variation mixes **exogenous** variation (price moves for its own reasons → the causal effect, downward) with **endogenous** variation (weather moves price *and* demand → upward). OLS on the mix gives a slope that is neither. Adjustment and experiments are ways to isolate the exogenous part.

Exogeneity

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$$

- The most important assumption when the goal is the **effect** of x on y
- The error term must not be related to x: $\mathrm{Cov}(x_i, \varepsilon_i) = 0$ — x is **exogenous**
- If $\mathrm{Cov}(x_i, \varepsilon_i) \neq 0$: x is **endogenous**, the estimate is **biased** (and stays biased with more data), and β_1 cannot be read as causal
- Randomization **guarantees** exogeneity. Opt-in **violates** it. Each adjustment variable moves a piece of ε_i into the model: `prior_hours` took most of the enthusiasm out (9.3 → 4.5); what is left still correlates with adoption (4.5 > 3)

Exogeneity cannot be tested. The residuals of a fitted regression are uncorrelated with the predictors *by construction* — the data always "look" exogenous. The argument has to come from outside: theory, and how the data came to be (who chose the treatment, and why).

Your turn (5 min)

Add **age** to the full model. Does the Smart Mix coefficient change? Why (not)?

```
model_age <- lm(hours ~ smart_mix + prior_hours + tenure + premium + device + age,
                data = smartmix_obs)
round(coef(summary(model_age)), 3)
```

Then: which variable in the data set should you **never** adjust for when estimating the effect of Smart Mix on listening — and why?

5.5 Categorical predictors

Dummies and the reference category

```
coef(model_full)[c("premium", "deviceAndroid", "deviceDesktop")]
```

```
premium deviceAndroid deviceDesktop
  2.924        −1.475        −2.488
```

- **0/1 variable** (`premium`): coefficient = difference between the two groups, other variables held constant
- **Several categories** (`device`): R picks a **reference** (first level: iOS) and creates one **dummy** per other category
- Each dummy = difference to the **reference**

Android users: **1.5 hours less** than iOS users; Desktop: **2.5 hours less** — holding everything else constant. *p*-values compare each category with the reference; “does device matter at all?” → `anova(model_without, model_full)` (the ANOVA of Chapter 4).

5.6 Prediction vs. explanation

Predicting listening time for new users

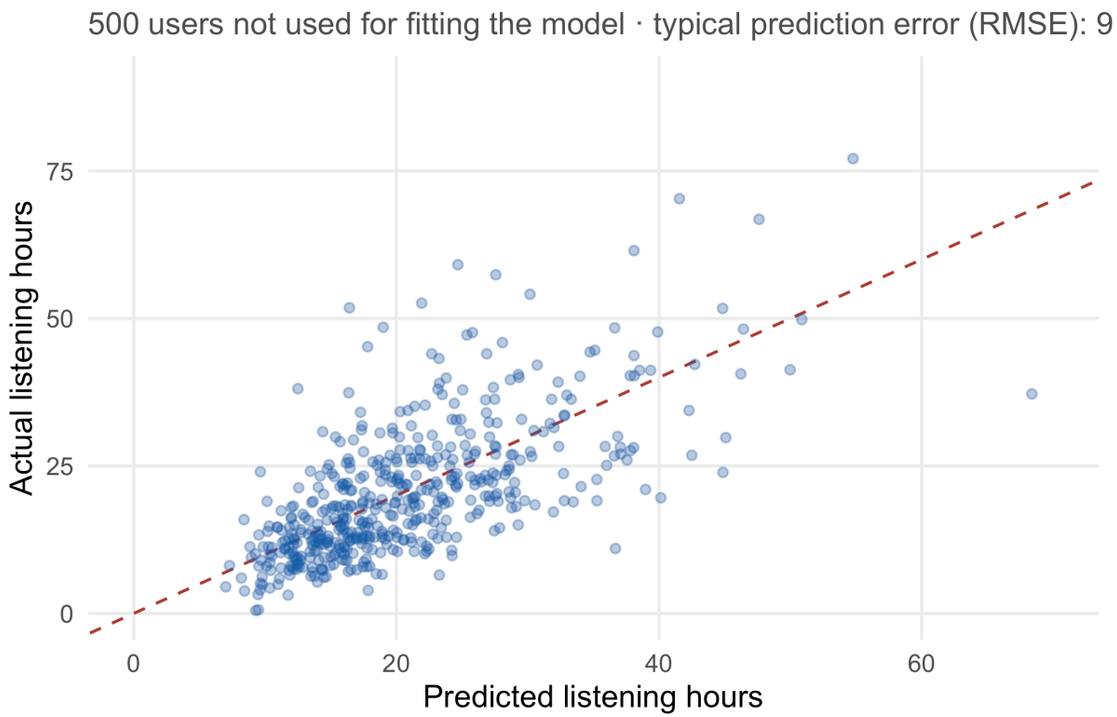
```r
set.seed(42)
train_ids <- sample(nrow(smartmix_obs), 1500)
train <- smartmix_obs[train_ids, ]
test  <- smartmix_obs[-train_ids, ]
model_pred <- lm(hours ~ prior_hours + tenure + premium +
                   device + smart_mix, data = train)
test$predicted <- predict(model_pred, newdata = test)

summary(model_pred)$r.squared        # in-sample R² (training data)
```

```
[1] 0.4817
```

```r
cor(test$predicted, test$hours)^2    # out-of-sample R² (test data)
```

```
[1] 0.4235
```



Fit on the training users, judge on the test users. The in-sample R^2 flatters the model (it was fitted to these very users); the **out-of-sample R^2** — the squared correlation between predicted and actual values in the test set — is the honest number. Close here (0.48 vs. 0.42): few predictors, many users. With many predictors and small samples, the gap is **overfitting** made visible.

How far off is a prediction? The RMSE

$$\text{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}$$

1. Each test user's **prediction error** $y_i - \hat{y}_i$
2. **Square** it: over- and under-predictions do not cancel, big misses count more
3. **Average** the squares
4. **Root**: back to the original unit (hours)

```
1  sqrt(mean((test$hours - test$predicted)^2))
```

```
[1] 9.425
```

The RMSE is the **standard deviation of the prediction errors**: a typical prediction for a new user misses by about **9.5 hours**. For a decision maker, that is more useful than a share of explained variation — it is in the unit of the decision.

Two questions, two rules

Explanation (causal question): include confounders, exclude mediators, state what is unobserved. The **coefficient** matters; R^2 is secondary.

Prediction: include whatever predicts; judge by **out-of-sample errors**. Coefficients are not interpreted; the **predictions** matter.

A model can be a **good predictor and a bad explainer** at the same time. Where in your company would each mistake — reading causal effects off a prediction model, or judging a causal model by its R^2 — cost money?

Coding block: two cases (40 min)

Case 1: podcast promotion — `podcast_promo.csv`, 400 episodes: promotion spend, host followers, downloads.

1. Scatter plot of downloads against spend; the **correlation**; one sentence
2. `lm()` + `summary()`: interpret the **slope**; read off **R^2**
3. Is the slope **causal**? What would you need to argue?

Case 2: a loyalty program — `loyalty_program.csv`, 1,500 customers, membership is **opt-in**.

1. **Naive:** `spending ~ member` — interpret; why not causal?
2. **Adjust** for `spend_prev`, `tenure`, `device`: how does the coefficient change, what does it mean now? With 95% CI
3. Interpret the coefficient of `device`
4. Is membership **exogenous** after adjustment? Argue from the data-generating process

Laptops open — alone or with your neighbor. Section "Coding block" in `05-observational-data.R`.

> At the end: one table presents case 1, another table case 2 — Menti first: enter your slope (case 1) and your adjusted coefficient (case 2).

Learning check (Part I)

1. Slope of hours on prior hours = 0.73. Interpret — and why is it not causal?
2. Adding a variable can increase, decrease, or leave the Smart Mix coefficient unchanged. When does each happen?
3. `planFamily` = 4.2 with reference “Free”. Interpret. How do you test whether the plan matters at all?
4. R^2 = 0.71 for predicting next month’s spending with 25 predictors on 300 customers. What do you ask?

Next session

Observational data II — beyond the straight line, and a quasi-experiment

- Functional form: when a line is wrong — **elasticities**
- Yes/no outcomes: logistic regression
- The three diagnostics that change decisions
- **Difference-in-differences** with real playlist data

> Before next session: work through §5.1–5.6 on the course site. **Debrief of Assignment 2** at the start of the session.

Part 2 · Observational data II

Before we start

Debrief A2 + quiz Chapter 5, Part I *(15 min, one sheet)*

- Page 1: debrief of **Assignment 2** (3 assignment points)
- Page 2: quiz on **Chapter 5, §5.1–5.6** (participation)
- Closed book, no laptop

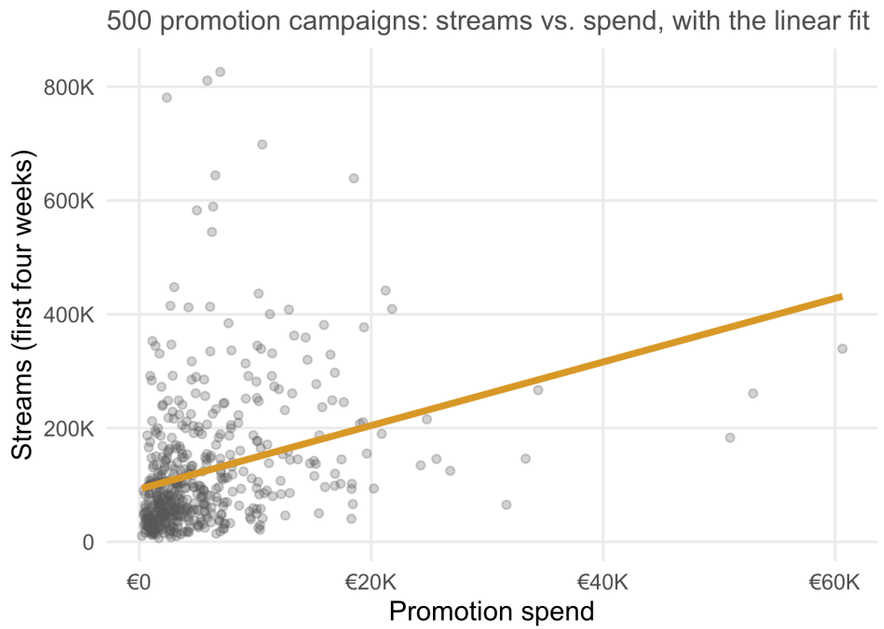
Assignment 3 is handed out today (end of session).

Devices only when we code — as last time

> Today: functional form and elasticities → logistic regression → diagnostics → difference-in-differences → **one coding block with two cases** → A3.

5.7 Functional form: when the line is wrong

Two assumptions of the linear model



The case. A label runs **500 promotion campaigns** for new releases; for each we know the **spend** (euros), the artist's followers, the playlist placements, the genre — and the **streams in the first four weeks**. The chart: streams against spend, one dot per campaign, with the linear regression line.

$$y_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + \cdots + \varepsilon_i$$

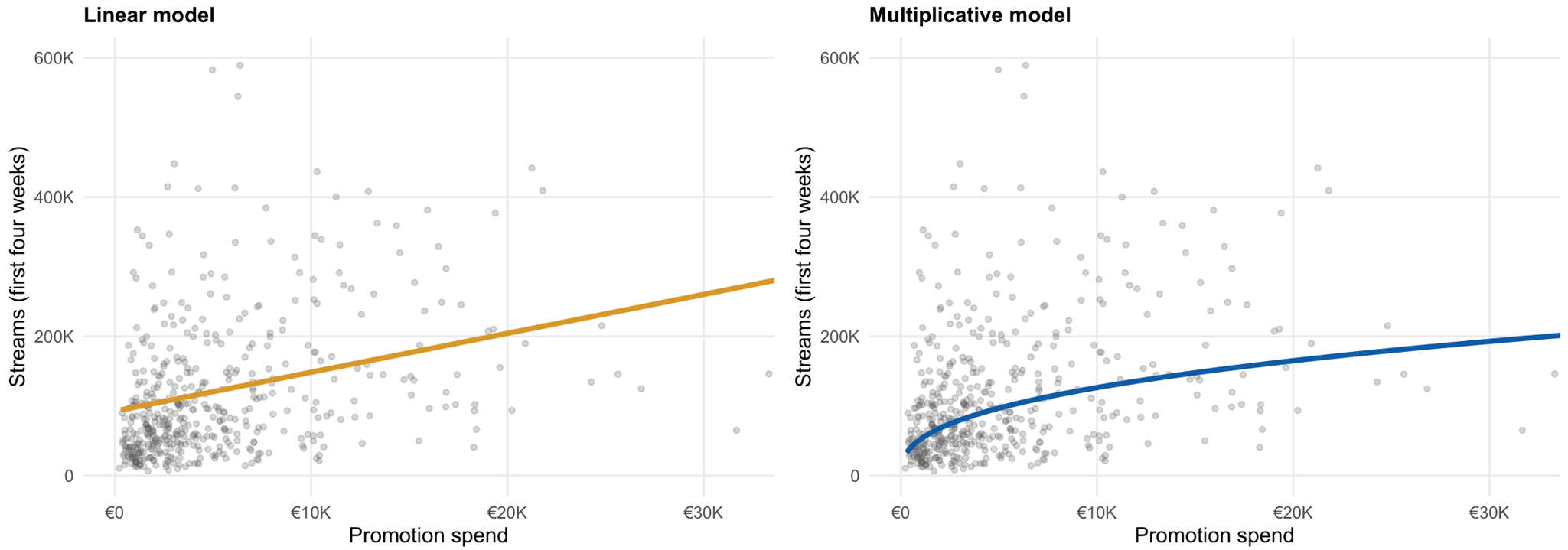
Linearity: a one-unit change in x has the same effect at every level of x — **constant marginal returns**: 10 → 11 € adds as many streams as 100,000 → 100,001 €. **Additivity:** the effect of one predictor does not depend on the others.

> **Relaxed where:** linearity → the **multiplicative (log-log) model** (this section) · additivity → the multiplicative model and **interactions** (Chapter 6) · **detecting** a wrong form → the residual plot (§5.9)

Two models for the same data

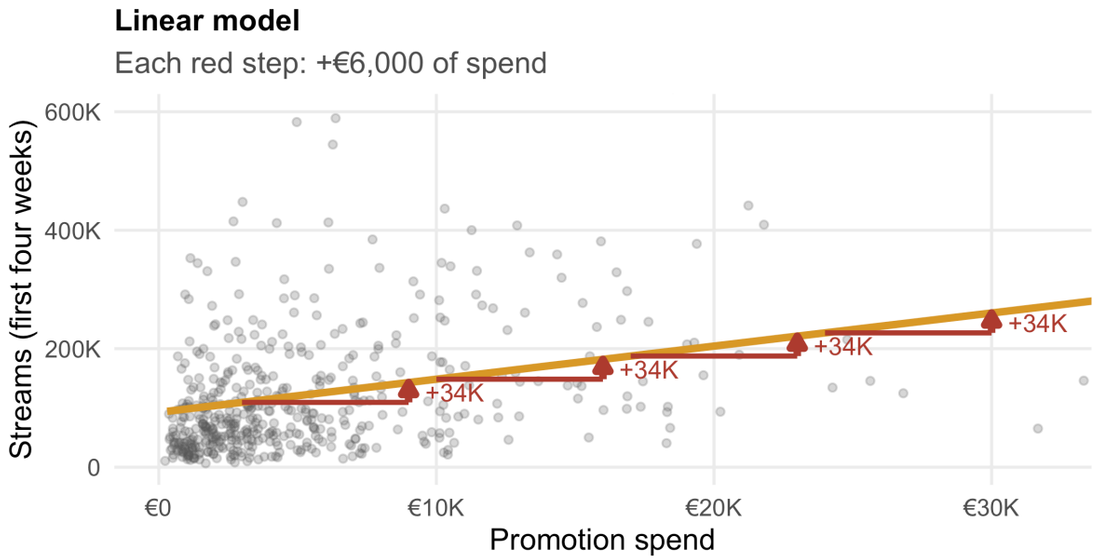
$$\text{streams}_i = \beta_0 + \beta_1 \, \text{spend}_i + \varepsilon_i$$

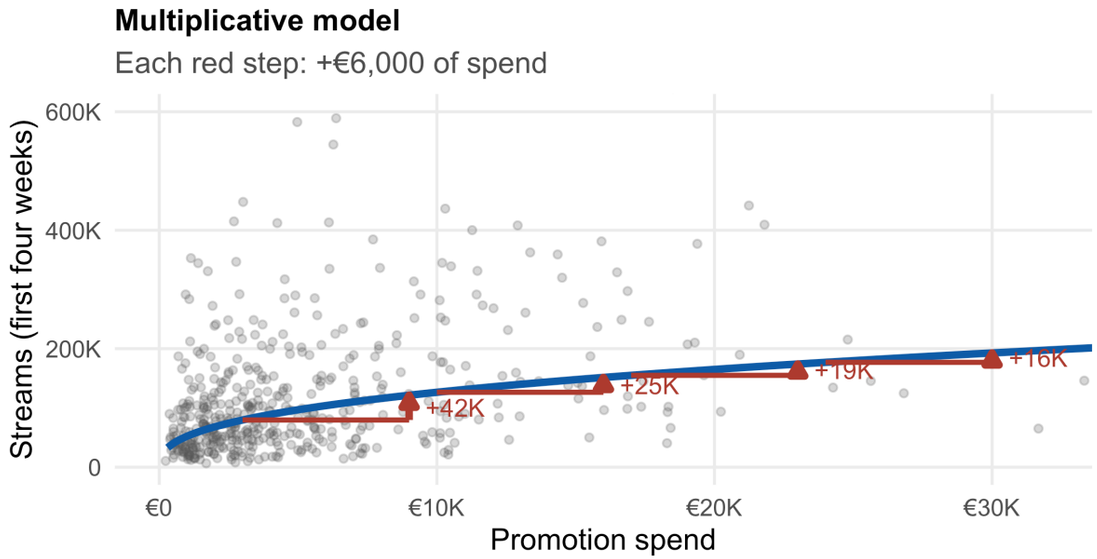
$$\text{streams}_i = \beta_0 \cdot \text{spend}_i^{\beta_1} \cdot \varepsilon_i$$



Left: the linear model — a straight line. **Right:** the **multiplicative** model — spend enters as a *power*, and the fitted curve bends: the first euros buy many streams, later euros fewer. Same data, same question, a different assumption about *how*

Marginal returns: what one more step buys





- **Constant marginal returns:** every extra €6,000 buys the **same** +34K streams
- **Additive:** spend adds a fixed number of streams on top of everything else
- **Reading β_1:** +1 unit of $x \rightarrow +\beta_1$ units of y (5.6 streams per euro)

- **Decreasing marginal returns:** +€6,000 buys +45K streams at a small budget, +12K at a large one
- **Multiplicative:** spend *scales* the streams — a **percentage** of what is already there
- **Reading β_1:** +1% of $x \rightarrow +\beta_1$% of y (0.38% per 1% spend: the **elasticity**)

Estimating the multiplicative model: take logs

Multiplicative: each predictor *scales* the outcome

$$\text{streams}_i = e^{\beta_0} \cdot \text{spend}_i^{\beta_1} \cdot \text{followers}_i^{\beta_2} \cdot e^{\varepsilon_i}$$

Take logs on both sides → additive again, a job for `lm()`:

$$\log(\text{streams}_i) = \beta_0 + \beta_1 \log(\text{spend}_i) + \beta_2 \log(\text{followers}_i) + \varepsilon_i$$

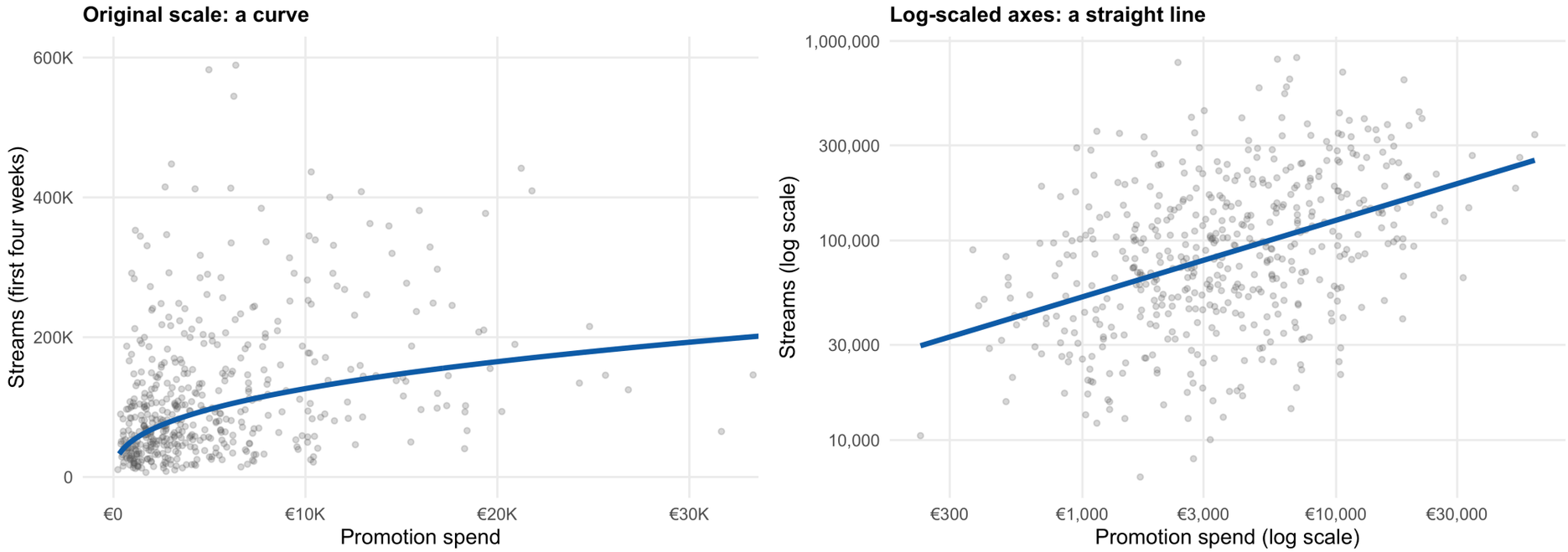
Three log rules do the work:

- $\log(a \cdot b) = \log a + \log b$ — a product becomes a sum
- $\log(a^b) = b \log a$ — a power becomes a factor
- $\log(e^x) = x$

Additive model: each euro adds β_1 streams, always.
Multiplicative model: each euro adds a **percentage** of the current streams — diminishing returns built in; and the effect of spend is *scaled* by the followers.

The same curve on log-scaled axes

$$\text{streams}_i = \beta_0 \cdot \text{spend}_i^{\beta_1} \cdot \varepsilon_i \qquad \log(\text{streams}_i) = \log \beta_0 + \beta_1 \log(\text{spend}_i) + \log \varepsilon_i$$



Left: the multiplicative model on the original scale — a curve with decreasing marginal returns. **Right:** the same data with **both axes log-scaled** (each gridline is ×10 the previous one; the units stay euros and streams): the curve becomes a **straight line**, and its slope is β_1. That is the model `lm(log(streams) ~ log(spend))` estimates.

Both models in R: the full output

$$\text{streams}_i = \beta_0 + \beta_1 \, \text{spend}_i + \varepsilon_i$$

```r
model_lin_simple <- lm(streams ~ spend, data = campaigns)
summary(model_lin_simple)
```

```
Call:
lm(formula = streams ~ spend, data = campaigns)

Residuals:
    Min      1Q  Median      3Q     Max
-204459  -67186  -34157   27654  694356

Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept) 92600.025   6782.260   13.65   <2e-16 ***
spend           5.589      0.781    7.16    3e-12 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 114000 on 498 degrees of freedom
Multiple R-squared:  0.0932,    Adjusted R-squared:  0.0914
F-statistic: 51.2 on 1 and 498 DF,  p-value: 3.01e-12
```

Reading: one more euro → **+5.6 streams**, at every budget; intercept 92,600 = predicted streams at zero spend.

$$\log(\text{streams}_i) = \beta_0 + \beta_1 \log(\text{spend}_i) + \varepsilon_i$$

```r
model_loglog_simple <- lm(log(streams) ~ log(spend), data = campaigns)
summary(model_loglog_simple)
```

```
Call:
lm(formula = log(streams) ~ log(spend), data = campaigns)

Residuals:
    Min      1Q  Median      3Q     Max
-2.2804 -0.5076  0.0152  0.5652  2.3720

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)   8.2098     0.3094    26.5   <2e-16 ***
log(spend)    0.3841     0.0375    10.2   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

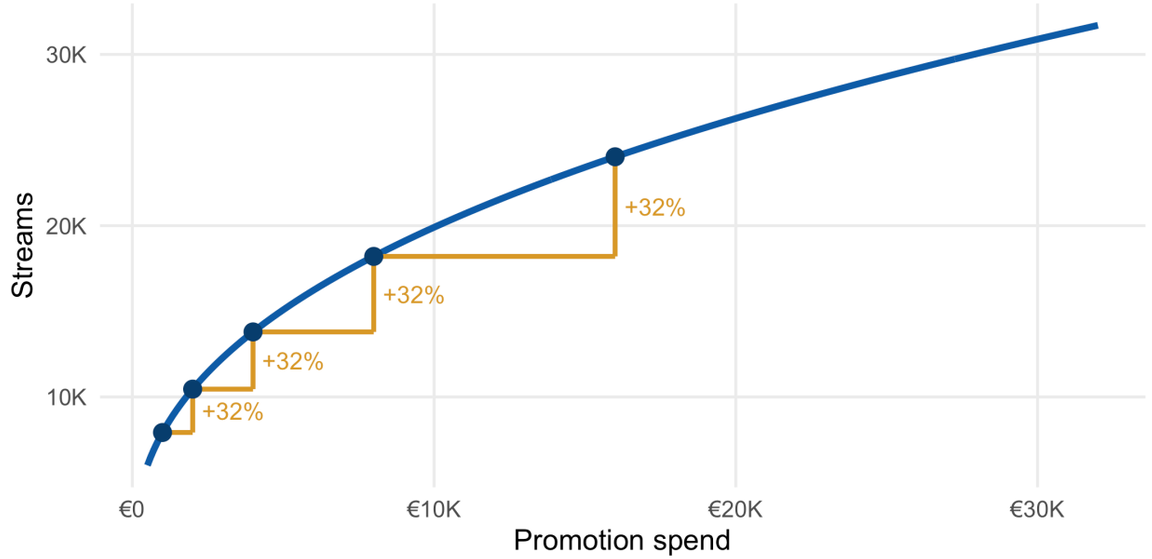
Residual standard error: 0.802 on 498 degrees of freedom
Multiple R-squared:  0.174, Adjusted R-squared:  0.173
F-statistic:  105 on 1 and 498 DF,  p-value: <2e-16
```

Reading: +1% spend → **+0.38% streams** (elasticity), at every budget. **R^2 0.17 vs. 0.09:** the log-log model explains more of the variation — a first hint that its shape fits the data better (the residual plots in §5.9 confirm it).

Why the log transformation straightens the curve

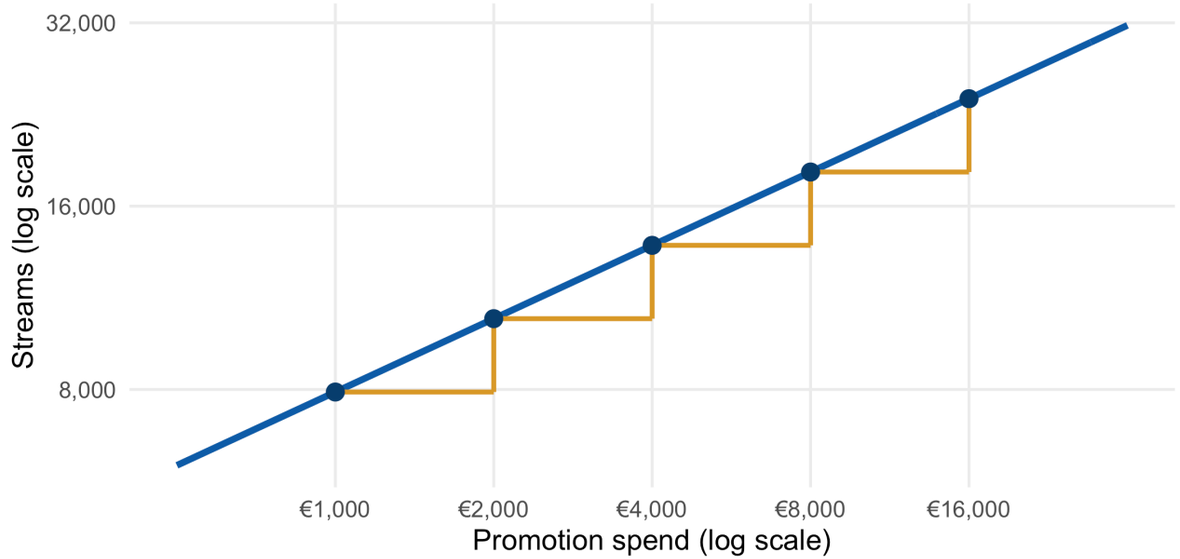
Original scale

Each doubling of spend adds the same percentage of streams, but a shrinking number of streams



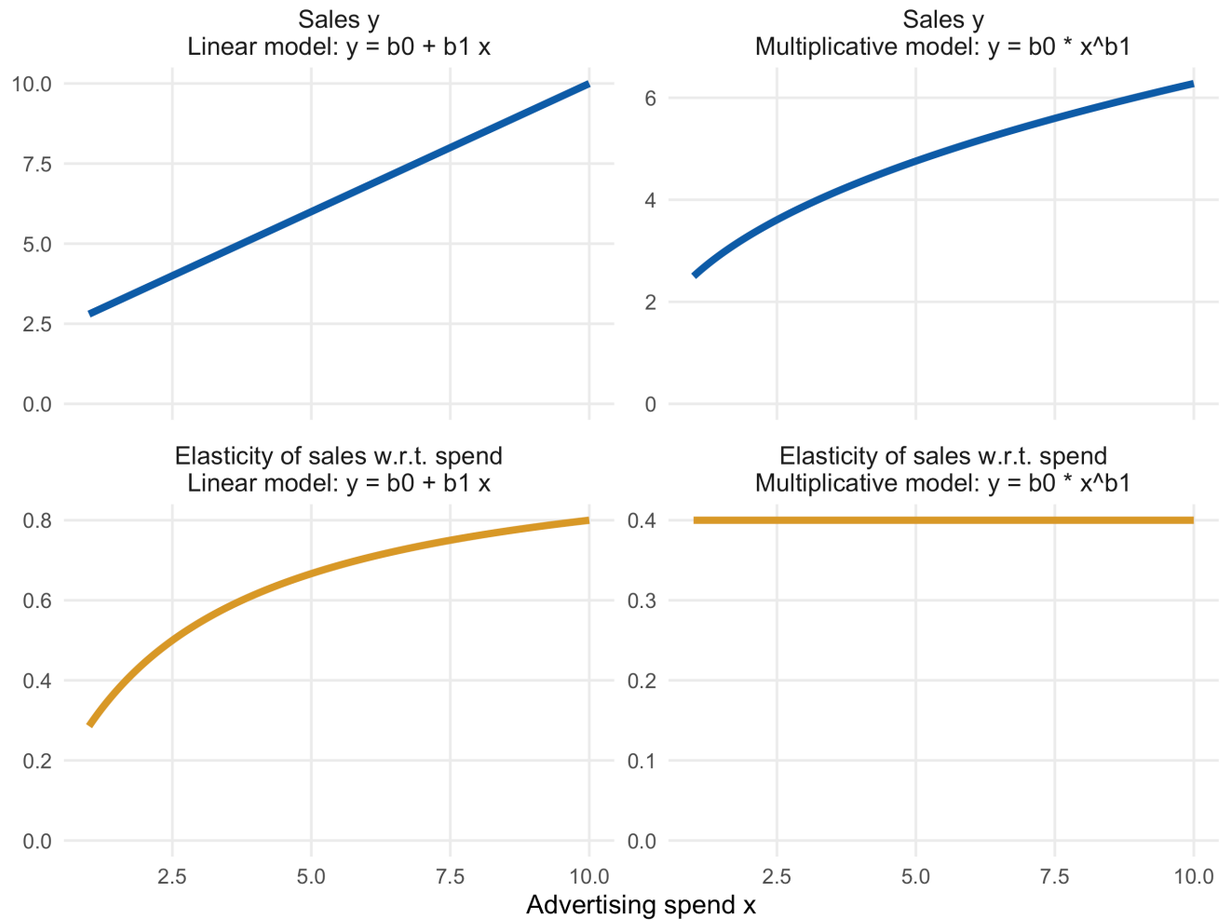
Log-scaled axes (same units)

Equal percentage steps become equal distances: the curve is a straight line with slope 0.4



Multiplicative model: equal **percentage** steps in x → equal **percentage** steps in y (each doubling of spend adds the same percentage of streams, but ever fewer streams). The logarithm turns percentages into distances — a doubling is always $\log 2$ — so the curve becomes a line with slope = elasticity.

Elasticity: linear vs. log-log model



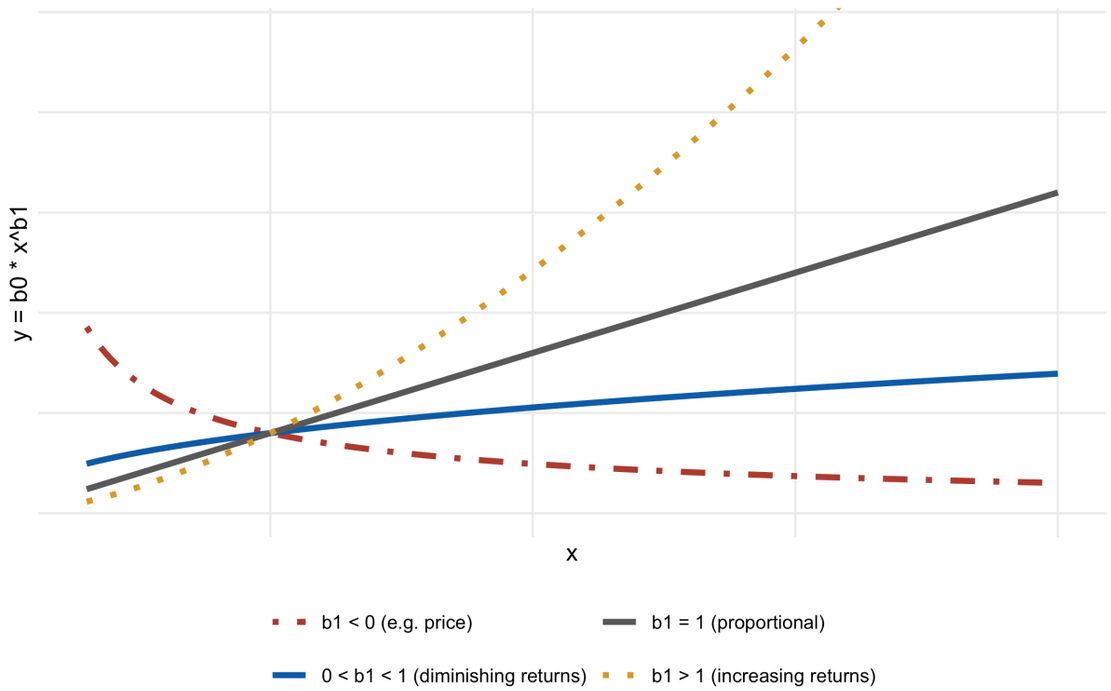
Elasticity $\eta_x = (\Delta y/y)/(\Delta x/x)$: the % change in y per % change in x

- **Linear:** $\eta_x = \beta_1\, x/y$ **rises** with x — advertising gets *relatively more* effective the more you spend: implausible
- **Multiplicative:** $\eta_x = \beta_1$ **constant**; the marginal returns fall

> Additive in **logs** = multiplicative in **levels**: the same *percentage* effect for every artist; the *number* of streams it buys scales with the follower base.

After Hanssens, Parsons & Schultz (2001), Market Response Models.

What the elasticity says about the shape



- $\beta_1 < 0$: falls, flattening out — **price**
- $0 < \beta_1 < 1$: **diminishing returns** — advertising, promotion; most marketing elasticities (< 0.5)
- $\beta_1 = 1$: **proportional** — a line through the origin; the one linear case
- $\beta_1 > 1$: **increasing returns** — each extra euro buys *more* than the last. Rare: **network effects** (a messaging app or marketplace gains value with every user), or a promotion that crosses a **threshold** and spreads by word of mouth

> Not defined at $x = 0$; no minimum level of y. Optimum: zero or infinity in the linear model, finite in the multiplicative one.

The full log-log model: an elasticity per input

```r
model_loglog <- lm(log(streams) ~ log(spend) + log(followers) + placements + genre, data = campaigns)
summary(model_loglog)
```

```
Call:
lm(formula = log(streams) ~ log(spend) + log(followers) + placements +
    genre, data = campaigns)

Residuals:
    Min      1Q  Median      3Q     Max
-1.5962 -0.2684  0.0284  0.3030  1.4799

Coefficients:
                  Estimate Std. Error t value Pr(>|t|)
(Intercept)         2.8815     0.2346   12.28  < 2e-16 ***
log(spend)          0.3310     0.0207   16.00  < 2e-16 ***
log(followers)      0.5071     0.0157   32.28  < 2e-16 ***
placements          0.1203     0.0119   10.07  < 2e-16 ***
genreHipHop/Rap     0.1994     0.0476    4.19 0.000033 ***
genreRock          -0.2555     0.0599   -4.27 0.000023 ***
genreElectro/Dance  0.1026     0.0594    1.73    0.085 .
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.44 on 493 degrees of freedom
Multiple R-squared:  0.754, Adjusted R-squared:  0.751
F-statistic:  251 on 6 and 493 DF,  p-value: <2e-16
```

`log(spend)` = **0.33**: +1% spend → +0.33% streams, at every budget. Elasticities have no units and are **comparable**: followers (0.51) matter more than the budget. `placements` = 0.12: a semi-elasticity, +12% per placement.

A **marketing mix model** in miniature: an **elasticity per channel** → shift budget to where a euro buys the largest percentage gain. Real MMMs add carryover and saturation — and spend is **endogenous**, so companies calibrate them with experiments.

Four ways to combine levels and logs

Model	Equation	Reading β_1	Typical use
Level-level	$y = \beta_0 + \beta_1 x$	+1 unit $x \to \beta_1$ units of y	Constant effects; dummies
Log-level	$\log y = \beta_0 + \beta_1 x$	+1 unit $x \to \approx 100\beta_1\%$ of y (exact: $100(e^{\beta_1} - 1)\%$): **semi-elasticity**	A placement adds 12%
Level-log	$y = \beta_0 + \beta_1 \log x$	+1% $x \to \approx \beta_1/100$ units of y	Diminishing returns, outcome in units
Log-log	$\log y = \beta_0 + \beta_1 \log x$	+1% $x \to \beta_1\%$ of y: **elasticity**	Spend, price: the marketing default

> The transformation decides the **unit of the coefficient**: units, percent, or percent per percent. In our model, `log(spend)` and `log(followers)` are elasticities; `placements` and the genre dummies are semi-elasticities (percent per unit).

Your turn (5 min)

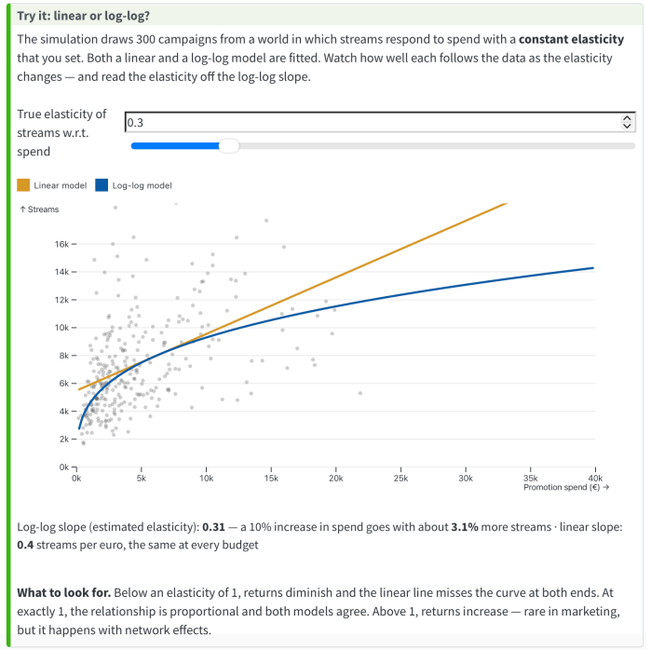
From the log-log model:

1. `genreHipHop/Rap` = 0.20. By what percentage do HipHop/Rap campaigns differ from Pop campaigns (the reference) — approximately and exactly?
2. A marketing manager proposes to raise the promotion budget of every campaign by **10%**. By what percentage would streams increase, according to the model? Is this a causal claim?

Live demo: linear or log-log?

Course book, §5.7 — move the **true elasticity**.

https://wu-rds.github.io/MA2026/05-observational-data.html#sec-obs-form



> Elasticity **< 1**: diminishing returns, the line misses at both ends. **= 1**: proportional. **> 1**: increasing returns (rare — network effects).

5.8 Yes/no outcomes: logistic regression

The case: who upgrades to Premium?

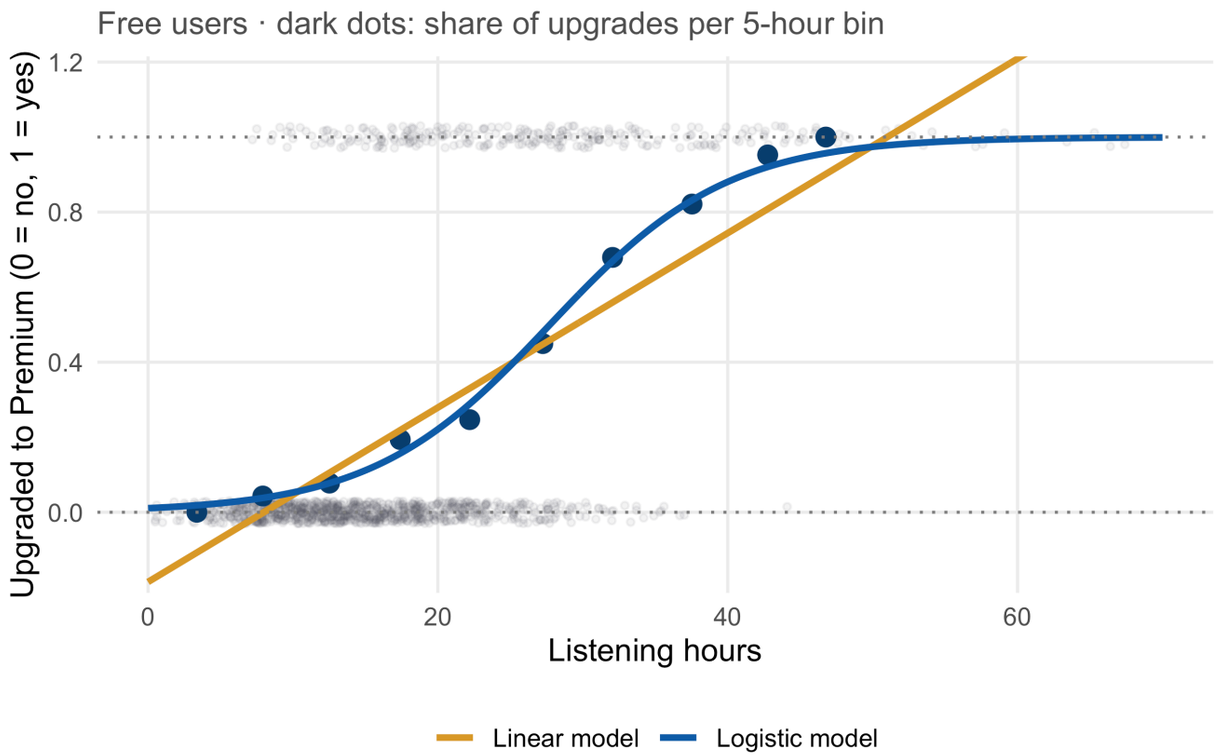
Same data as before — the Smart Mix launch — a new question. Among the **free users** (1,239 of the 2,000), the service recorded whether a user **upgraded to Premium** within three months after the launch.

- Outcome: `upgraded` — **1** if the user upgraded, **0** if not; 22% did
- Question of the growth team: *does listening time predict who upgrades — and how strongly?*
- The answer feeds a **targeting** decision: which free users should get the upgrade offer first?

A **yes/no outcome**. Everything so far modeled a *number* (hours, streams). Now the outcome is a **probability** — and a probability has to stay between 0 and 1.

What goes wrong if we just run `lm(upgraded ~ hours)`?

Why not a straight line?



- Outcome: **upgraded to Premium** within 3 months — free users, 22% did
- A line predicts probabilities **below 0** and **above 1** — and misses the shape
- The share of upgrades rises slowly, then steeply, then flattens near 100%: an **S-curve**

> We need a function that only takes values between 0 and 1.

The logistic function

$$P(y_i = 1) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x_i)}}$$

- $\beta_0 + \beta_1 x_i$ is the familiar line — fed through a function that squeezes it into [0, 1]: line far **below** 0 → $P \approx 0$ · line = 0 → P = 0.5 · line far **above** 0 → $P \approx 1$
- Derived from the **log-odds**: odds = $P/(1-P)$ ("1 to 4" = 0.25)

$$\log\left(\frac{P(y_i = 1)}{1 - P(y_i = 1)}\right) = \beta_0 + \beta_1 x_i$$

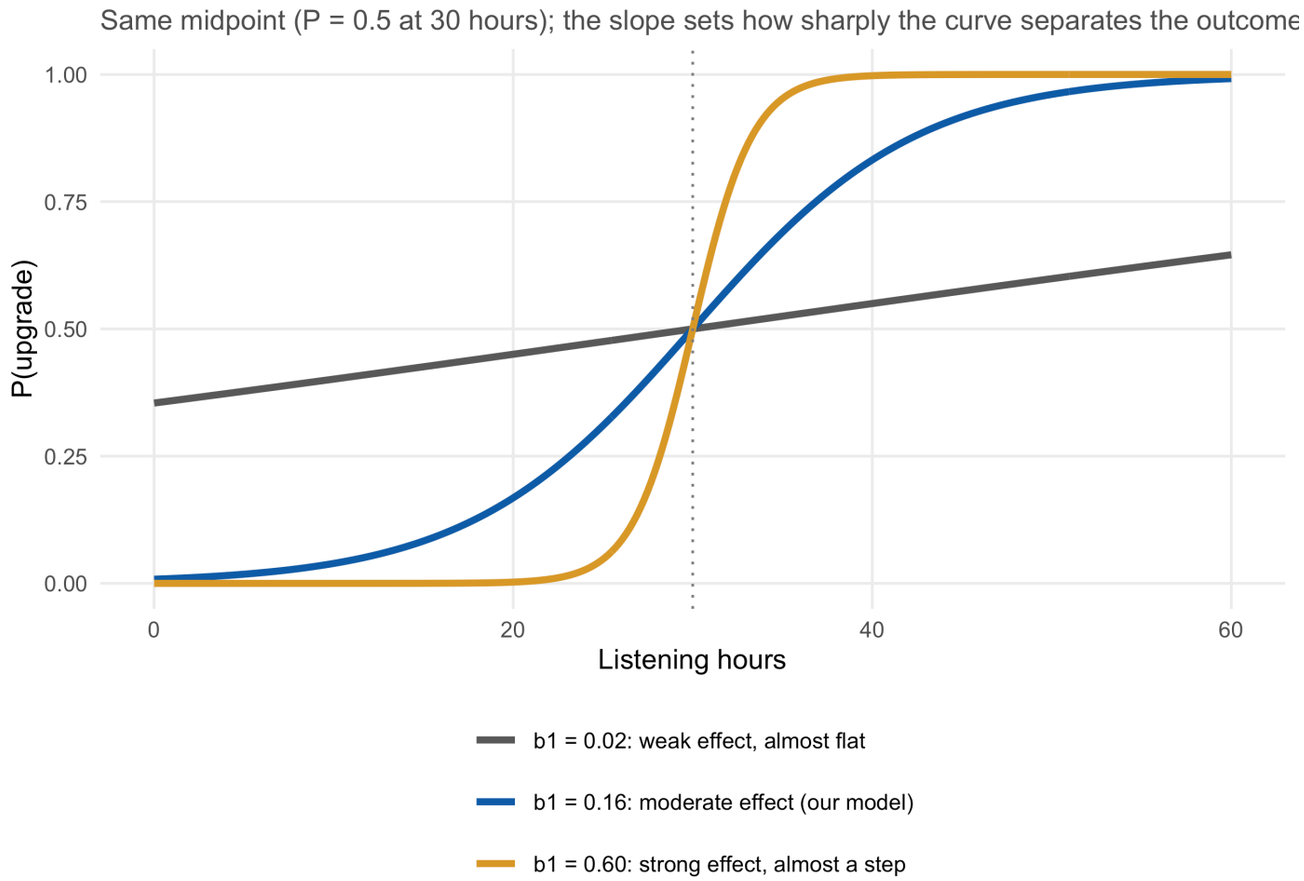
- Exponentiate → the odds: $\frac{P(y_i = 1)}{1 - P(y_i = 1)} = e^{\beta_0 + \beta_1 x_i}$

```r
free_users <- filter(smartmix_obs, premium == 0)
model_upgrade <- glm(upgraded ~ hours, family = binomial,
                     data = free_users)
round(coef(summary(model_upgrade)), 3)
```

```
            Estimate Std. Error z value Pr(>|z|)
(Intercept)   -4.517      0.244  -18.53        0
hours          0.163      0.011   15.05        0
```

> `glm()` = *generalized* linear model: the same predictors, a different link to the outcome. Estimated by **maximum likelihood**, not least squares (no line, no residuals to square): the coefficients under which the observed 0/1 data are most likely.

Weak vs. strong effects: the slope



- β_1 **≈ 0**: almost flat — the predictor hardly **discriminates** between upgraders and non-upgraders
- β_1 **large**: an almost vertical step — listening time nearly determines who upgrades
- β_0 shifts the curve; **midpoint** (P = 0.5) at $x = -\beta_0 / \beta_1$ **≈ 28 hours**

> **Sign** = direction, **size** = how sharply the outcome responds — on the log-odds scale; "how much" in probabilities: next two slides.

Reading the coefficients: odds ratios

```
1 exp(coef(model_upgrade))
```

```
(Intercept)       hours
    0.01093     1.17689
```

```
1 exp(confint(model_upgrade))
```

```
                2.5 %  97.5 %
(Intercept) 0.006661 0.01734
hours       1.152949 1.20296
```

- Coefficient 0.163 = change in the **log-odds** per hour — no intuition
- $e^{0.163}$ = **1.18**: each hour multiplies the **odds** of an upgrade by 1.18 (+18%) — the same factor everywhere
- Read **sign** and **significance** directly from the output

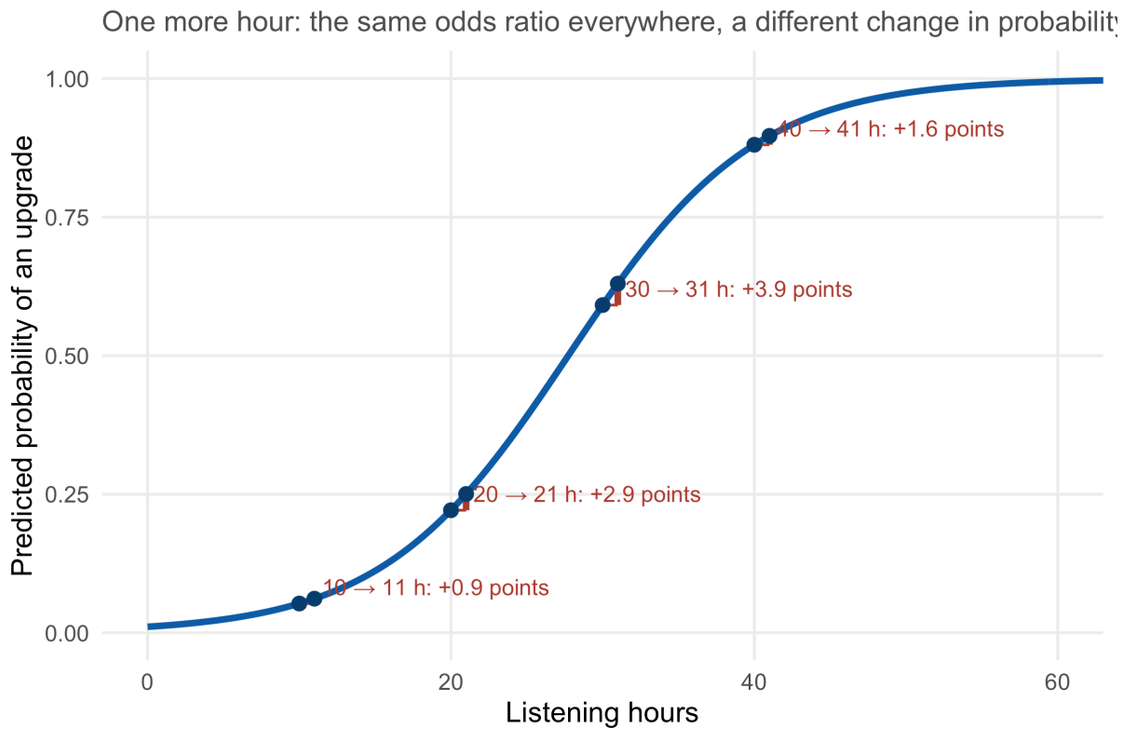
> **Odds ratio ≠ probability ratio.** "Odds +18%" is not "18% more likely" — unless the event is rare. For managers, translate into **probabilities**.

Reading the coefficients: probabilities at x and x + 1

```
predict(model_upgrade, type = "response",
        newdata = data.frame(hours = c(10, 11, 20, 21, 30, 31, 40, 41)))
```

```
      1       2       3       4       5       6       7       8
0.05275 0.06151 0.22111 0.25043 0.59135 0.63005 0.88062 0.89671
```

- 10 → 11 hours: **+0.9** points · 20 → 21: **+2.9** · 30 → 31: **+3.9** · 40 → 41: **+1.6**
- Same coefficient, same odds ratio — a **different change in probability**, largest around 50%



> Report effects as **predicted probabilities at chosen values** (10 vs. 30 hours: 5% vs. 59%) or as the **average change** across all users — “the effect of one more hour” has no single answer in percentage points.

A second case: who cancels?

Churn. For all 2,000 users, the data record whether the user **cancelled** the service within three months after the launch: `churn` = 1 (17% did), 0 otherwise.

- Question of the retention team: *which users are at risk — and does Premium status protect against churn, once listening and tenure are taken into account?*
- Candidates: `hours`, `premium`, `tenure`, `smart_mix` — the same predictors as in the adjustment models, now for a yes/no outcome
- The answer feeds a **retention** decision: whom to contact before they leave

> Logistic regression with **several predictors**: the same logic as multiple regression — each coefficient holds the others constant — but on the **log-odds** scale.

Several predictors: churn

```
model_churn <- glm(churn ~ hours + premium + tenure + smart_mix, family = binomial, data = smartmix_obs)
round(coef(summary(model_churn)), 3)
```

```
            Estimate Std. Error z value Pr(>|z|)
(Intercept)    0.277      0.161   1.717    0.086
hours         -0.061      0.009  -7.130    0.000
premium       -0.533      0.153  -3.489    0.000
tenure        -0.020      0.004  -5.452    0.000
smart_mix     -0.256      0.162  -1.579    0.114
```

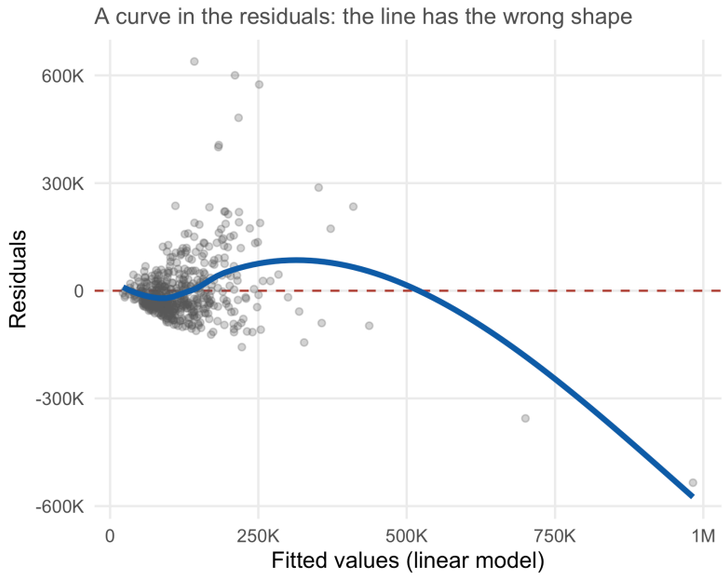
```
p_free    <- predict(model_churn, newdata = mutate(smartmix_obs, premium =
p_premium <- predict(model_churn, newdata = mutate(smartmix_obs, premium =
mean(p_premium - p_free)
```

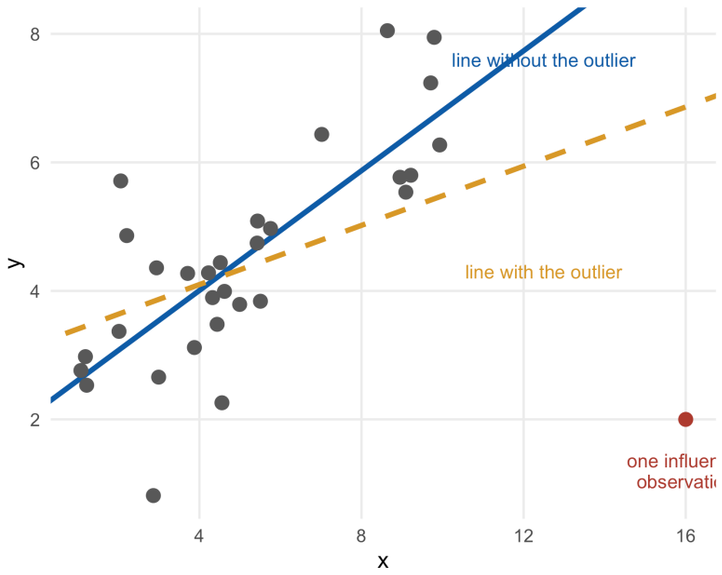
```
[1] -0.06477
```

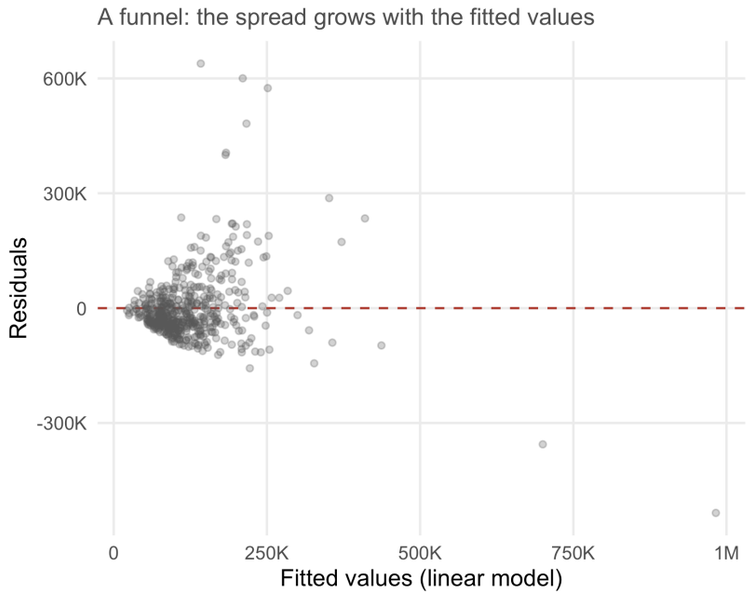
Same logic as multiple regression: each coefficient holds the others constant. Premium subscribers: about **9 percentage points** less likely to churn than otherwise identical free users. Causal? Exogeneity, again.

5.9 The diagnostics that change decisions

Three checks worth doing every time







1. Is the line the right shape? Curved residuals → change the functional form (log) or add a predictor.

2. Does one observation drive the result? `plot(model, which = 5)`; refit without it — if the conclusion changes, say so.

3. Is the uncertainty right? A funnel (test: `bptest(model)`) → **robust standard errors**: `coeftest(model, vcov = vcovHC(model))`.

5.10 A quasi-experiment: difference-in-differences

The case: what is a playlist listing worth?

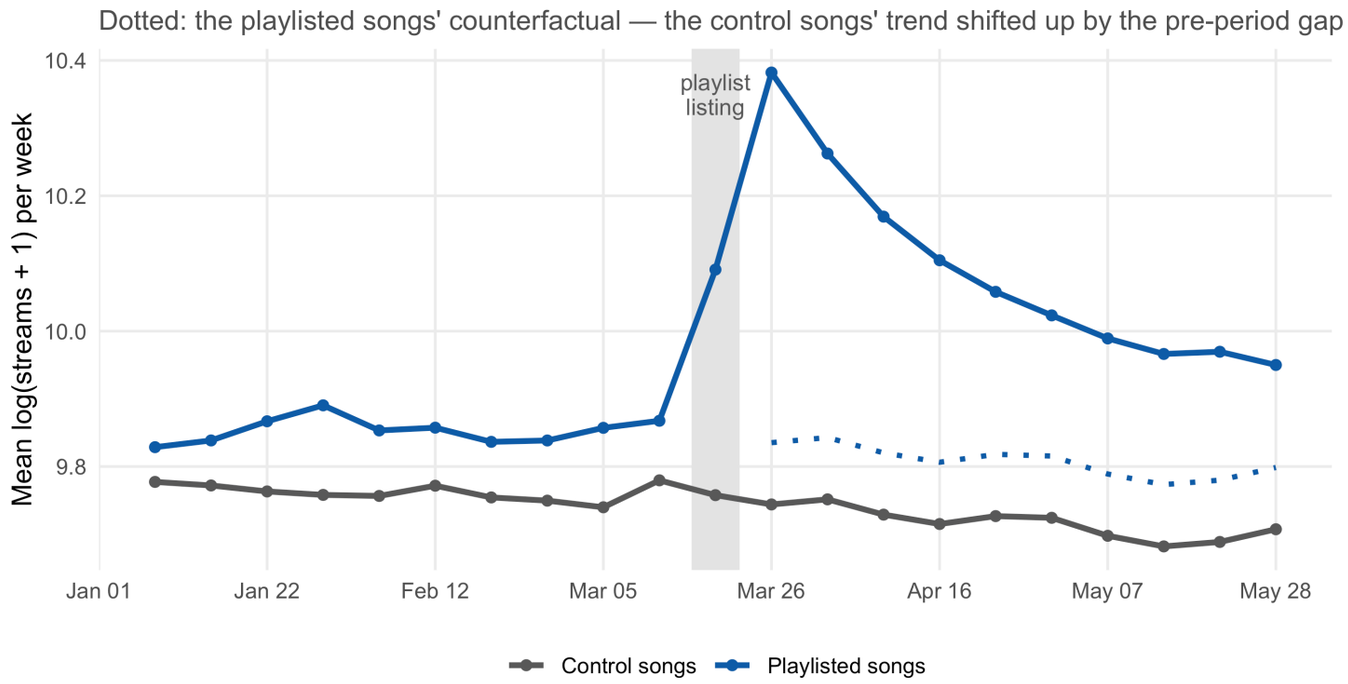
Real data from a WU research project. Labels pay a lot of attention to **editorial playlists**: being listed on a major playlist is supposed to boost a song's streams. But by how much?

- **290 songs**, weekly streams over **21 weeks** (January to June 2018)
- On **March 19, 2018**, 53 of them were added to a major playlist (the **treated** songs); the other 237 were not (the **control** songs)
- Question: *how much did the listing increase the treated songs' streams?*

No randomization: the curators chose the songs — and they pick songs that are already doing well. A before/after comparison of the treated songs alone would mix the listing effect with the general trend; a treated/control comparison would mix it with the curators' selection.

What do we need to isolate the effect of the listing?

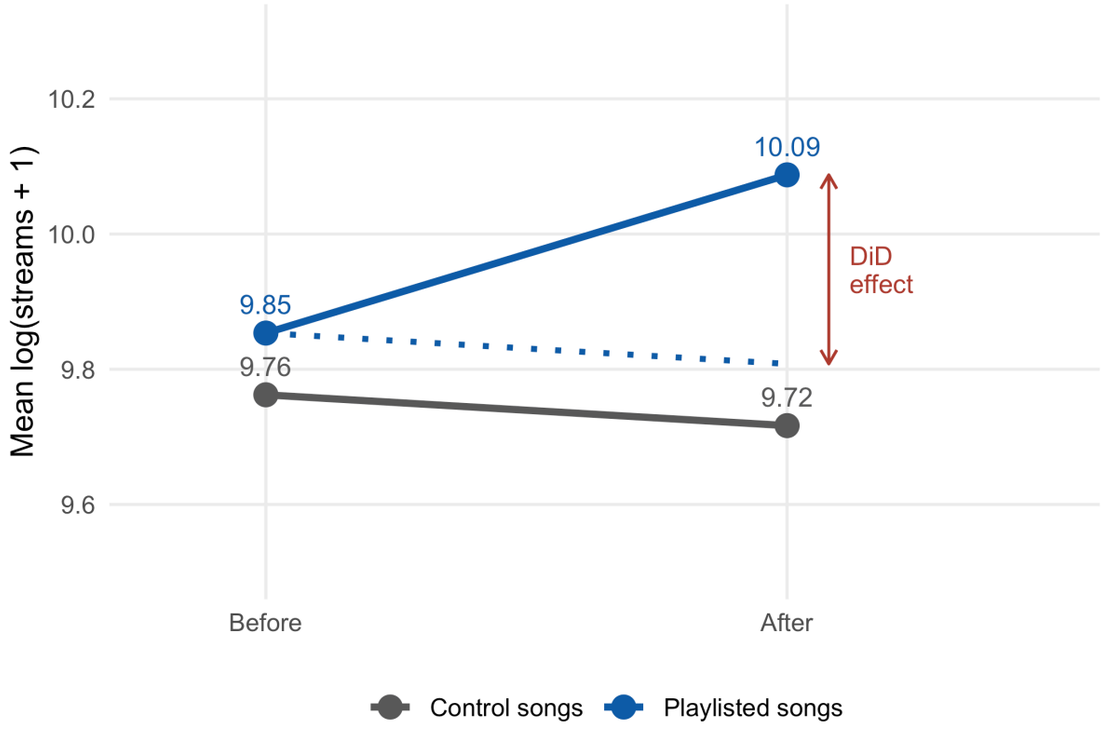
Playlist listings as a natural experiment



- **Real data:** 290 songs × 21 weeks
- March 19, 2018: **53 songs** added to a major playlist; 237 control songs
- Treated songs start **higher** → levels not comparable
- Before the listing: both groups move in **parallel**
- At the listing: the treated group **jumps**

Data: WU-RDS research project (github.com/WU-RDS/did).

Two differences, one estimate



$$\text{DiD} = (\bar{y}_{T,\text{after}} - \bar{y}_{T,\text{before}}) - (\bar{y}_{C,\text{after}} - \bar{y}_{C,\text{before}})$$

(10.09 − 9.85) − (9.72 − 9.76) = **0.28** log points

> The control group's change estimates what would have happened to the treated songs **without** the listing — the counterfactual (dotted).

The DiD regression: the model

$$\log(\text{streams}_{it}) = \beta_0 + \beta_1\,\text{treated}_i + \beta_2\,\text{post}_t + \beta_3\,(\text{treated}_i \times \text{post}_t) + \varepsilon_{it}$$

i = song, t = week. What the model predicts for the four cells:

	Before (post = 0)	**After (post = 1)**	**After − before**
Control (treated = 0)	β_0	$\beta_0 + \beta_2$	β_2 — the common time trend
Treated (treated = 1)	$\beta_0 + \beta_1$	$\beta_0 + \beta_1 + \beta_2 + \beta_3$	$\beta_2 + \beta_3$
Treated − control	β_1 — the level difference	$\beta_1 + \beta_3$	β_3 — **the DiD effect**

> β_3 is the difference of the two differences: the change of the treated songs **minus** the change of the control songs. In R, `treated * post` expands to the three terms.

The DiD regression: the estimate

```r
did_data <- playlist |>
  filter(week != as.Date("2018-03-19")) |>          # drop the listing week
  mutate(log_streams = log(streams + 1))
model_did <- lm(log_streams ~ treated * post, data = did_data)
round(coef(summary(model_did)), 3)
```

```
            Estimate Std. Error t value Pr(>|t|)
(Intercept)     9.762      0.018 545.865    0.000
treated         0.091      0.042   2.182    0.029
post           -0.046      0.025  -1.803    0.071
treated:post    0.280      0.059   4.727    0.000
```

Coefficient	Meaning
`(Intercept)` 9.76	Control, before: baseline
`treated` 0.09	Treated, before: the **level difference**
`post` −0.05	Control, after − before: the **common time trend**
`treated:post` **0.28**	The extra change of the treated songs: the **DiD effect** → ≈ **+32% streams**

The assumption: parallel trends

```r
placebo_data <- did_data |>
  filter(post == 0) |>
  mutate(fake_post = ifelse(week >= as.Date("2018-02-12"), 1, 0))
round(coef(summary(lm(log_streams ~ treated * fake_post,
                      data = placebo_data)))["treated:fake_post", ], 3)
```

```
Estimate Std. Error    t value   Pr(>|t|)
   0.002      0.085      0.027      0.978
```

> Without the listing, the treated songs would have followed the **same trend** as the controls. **Untestable** — but the pre-period makes it plausible: a fake listing in the pre-period shows **no effect**.

Live demo: course book, §5.10

https://wu-rds.github.io/MA2026/05-observational-data.html#sec-obs-did

Your turn (5 min)

Add `major_label` as a control variable to the DiD regression:

```
summary(lm(log_streams ~ treated * post + major_label, data = did_data))
```

Does the effect change? Then: name one situation in which the parallel-trends assumption would **fail** for playlist listings.

Coding block: two cases (40 min)

Case 3: podcast promotion, revisited — `podcast_promo.csv` again.

1. Scatter plot with **both axes logged**: what has changed?
2. **Log-log model** on spend and host followers; interpret both **elasticities**: what does +10% spend do?
3. Compare **R^2** with the linear model; which model for the budget advice?

Laptops open — alone or with your neighbor. Section "Coding block" in `05-observational-data.R`.

Case 4: winning back users — `app_reinstall.csv`, 1,200 users, **randomized** discount 0–60%, reinstalled (1/0), prior sessions.

1. **Share of reinstalls per discount level**, plotted: describe the shape
2. **Logistic regression** of `reinstalled` on `discount` and `prior_use`: sign and **odds ratio** of the discount
3. Predicted probabilities for 10 prior sessions at **10, 11, 40, 41%**: points per extra point of discount — why do the two answers differ?

> At the end: one table presents case 3, another case 4 — Menti first: your spend elasticity (case 3) and your odds ratio (case 4).

What observational data can and cannot tell you

Two strategies, one honest report

- **Adjustment** handles confounders you can **measure** — only as good as your proxies; “enough” is untestable
- **Difference-in-differences** handles confounders **constant over time** — needs parallel trends, plausible but never proven
- **Prediction** needs no causal assumptions — a good predictor is valuable even if no coefficient is an effect

> Report effect + uncertainty + **assumptions**: “Adjusting for prior listening, tenure, Premium and device, adopters listen ≈ 4.5 hours more (95% CI 3.6–5.3), assuming no further differences affect listening — the experiment (3.2) suggests this is not fully met.”

Assignment 3: analyzing observational data

- Your **personal** data (based on your student ID): an opt-in feature, promotion campaigns, and a small panel with a listing
- Tasks: naive vs. adjusted · categorical predictor · elasticity · logistic regression · DiD
- **Deadline:** Group A Sun, Nov 22 · Group B Wed, Nov 25
- **Debrief** at the start of the following session

> **10 points:** report 7 + debrief 3. State your **assumptions** — the best answers say what the data *cannot* show.

Learning check (Part II)

1. Log-log coefficient of ad spend = 0.15. Interpret. The level model says 0.8 sales per euro — why might the two disagree about a large budget increase?
2. Log-level model, dummy “promotion week” = 0.35. Percentage difference in sales — approximate and exact?
3. Residual plot shows a funnel. Consequence for coefficients, for *p*-values — and the simplest fix?
4. A DiD of a price promotion in treated stores: which assumption, and how can the pre-period help?

Next session

Module 6 · Mechanisms & heterogeneity — *Why, and for whom, does it work?*

- Interaction effects: when the effect differs between groups (moderation)
- Mediation: through which mechanism an effect works
- In experiments and observational data — and in your group project